

Spatial diffusion of innovation – agents of change, networks, spacial life cycles

Economic Complexity, Class 10 – Dec 2 2025

Social Data Science, Corvinus University of Budapest

Panel 3: shocks, diffusion, recent EC

8	Nov 18	Presentation of students' research proposal		
9	Nov 25	Gergő Tóth and Balázs Lengyel Shocks and adaptation: network resilience, systemic risks	III. Recent research	[13,1
10	Dec 2	Balázs Lengyel Spatial diffusion of innovation: agents of change, interregional networks, and spatial life cycles		[15,1
11	Dec 9	Viktor Stojkoski Digital trade and Multidimensional Economic Complexity		[17,1
12	Dec 16	Written exam / Presentation of Projects		

[13] Tóth, G., Elekes, Z., Whittle, A., Lee, C., & Kogler, D. F. (2022). Technology network structure conditions the economic resilience of regions. *Economic Geography*, 98(4), 355-378.

[14] Diem, C., Borsos, A., Reisch, T., Kertész, J., & Thurner, S. (2022). Quantifying firm-level economic systemic risk from nation-wide supply networks. *Scientific Reports*, 12(1), 7719.

[15] Elekes, Z., Boschma, R., & Lengyel, B. (2019). Foreign-owned firms as agents of structural change in regions. *Regional Studies*, 53(11), 1603-1613.

[16] Lengyel, B., Bokányi, E., Di Clemente, R., Kertész, J., & González, M. C. (2020). The role of geography in the complex diffusion of innovations. *Scientific Reports*, 10(1), 15065.

[17] Stojkoski, V., Koch, P., Coll, E., & Hidalgo, C. A. (2024). Estimating digital product trade through corporate revenue data. *Nature Communications*, 15(1), 5262.

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The Role of Spatial Effects in Regional Concentration

(Giuseppe Arbia, 2001)

- Key Issue: Traditional concentration measures (e.g., Gini) ignore geography.
- Core Proposition: Regional concentration is not only explained by a-spatial variability, but spatial clustering as well, where:
 - A-spatial measures focus on concentration, irrespective of location.
 - Spatial measures capture neighbor effects and polarization.
- Methodology: Italian manufacturing employment by province (1991) analyzed using:
 - Gini index (a-spatial)
 - Moran's I (spatial autocorrelation)
 - Getis-Ord index (local clustering)
- Findings: Measures are weakly correlated → each reveals different patterns.
- Conclusion: Combine spatial and a-spatial metrics for full understanding of concentration.

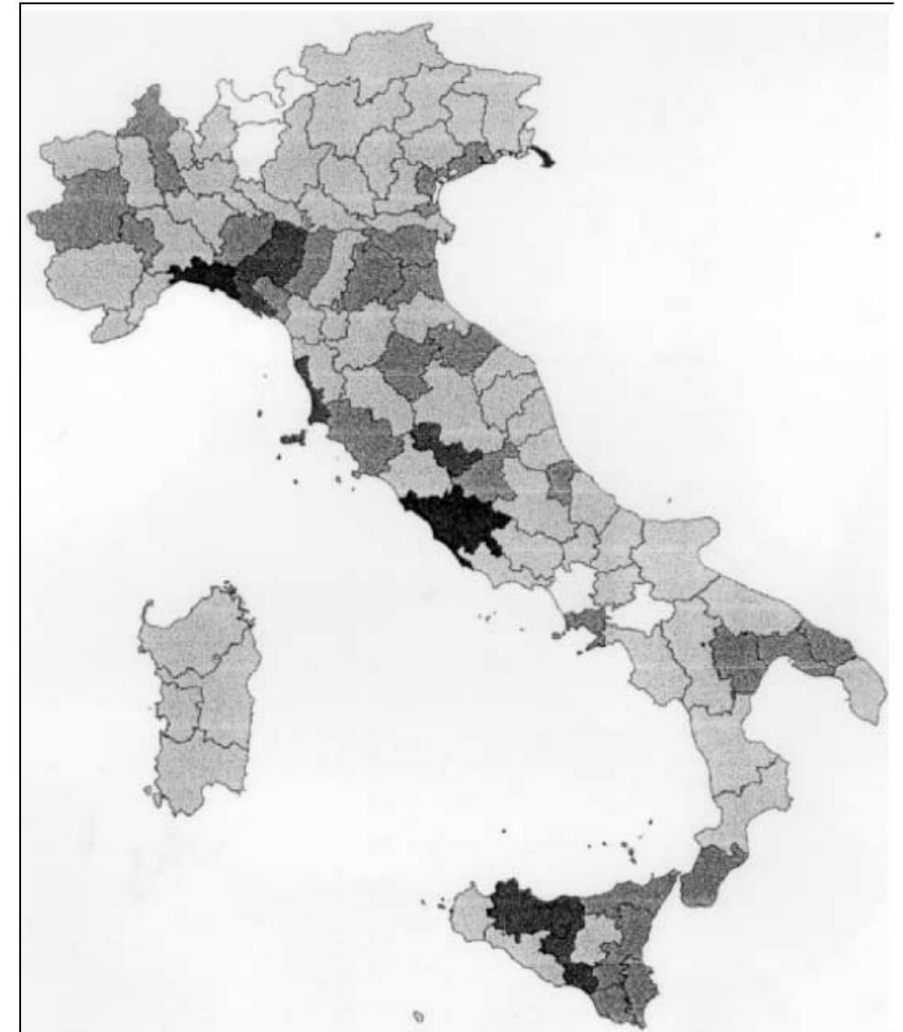
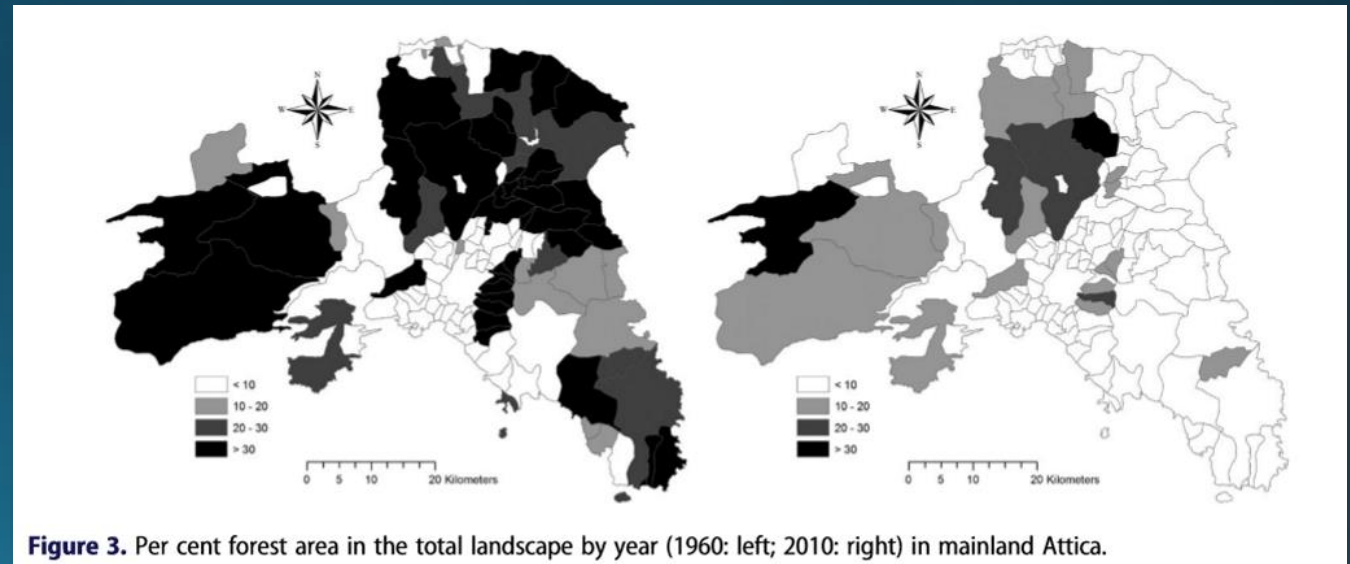


Fig. 5. Map of concentration of manufacturing employment in Italy in 1991. Grey-scale colours refer to the Gini location index. There are 5 classes: class 1 (*white*) refers to low concentration with G ranging from 0 to 0.1; class 2 refers to G ranging from 0.1 to 0.4; class 3 refers to G ranging from 0.4 to 0.6; class 4 refers to G ranging from 0.6 to 0.8 and class 5 (*black*) refers to G ranging from 0.8 to 1

Back to Von Thünen: A Southern European Perspective on Mono-centric Urban Growth

(Colantoni et al., 2017 – International Planning Studies)

- How does monocentric urban growth affect land use and natural areas in Southern Europe?
- Data:
 - Attica region, 1960-2010
 - 26 socioeconomic, environmental & land-use indicators
- Methods:
 - Statistical analysis
 - Correlation
 - Multiway factor analysis
- Key findings:
 - Urban expansion reduces forest and cropland near the core.
 - Planning tools can limit land loss



Defining urban clusters to detect agglomeration economies

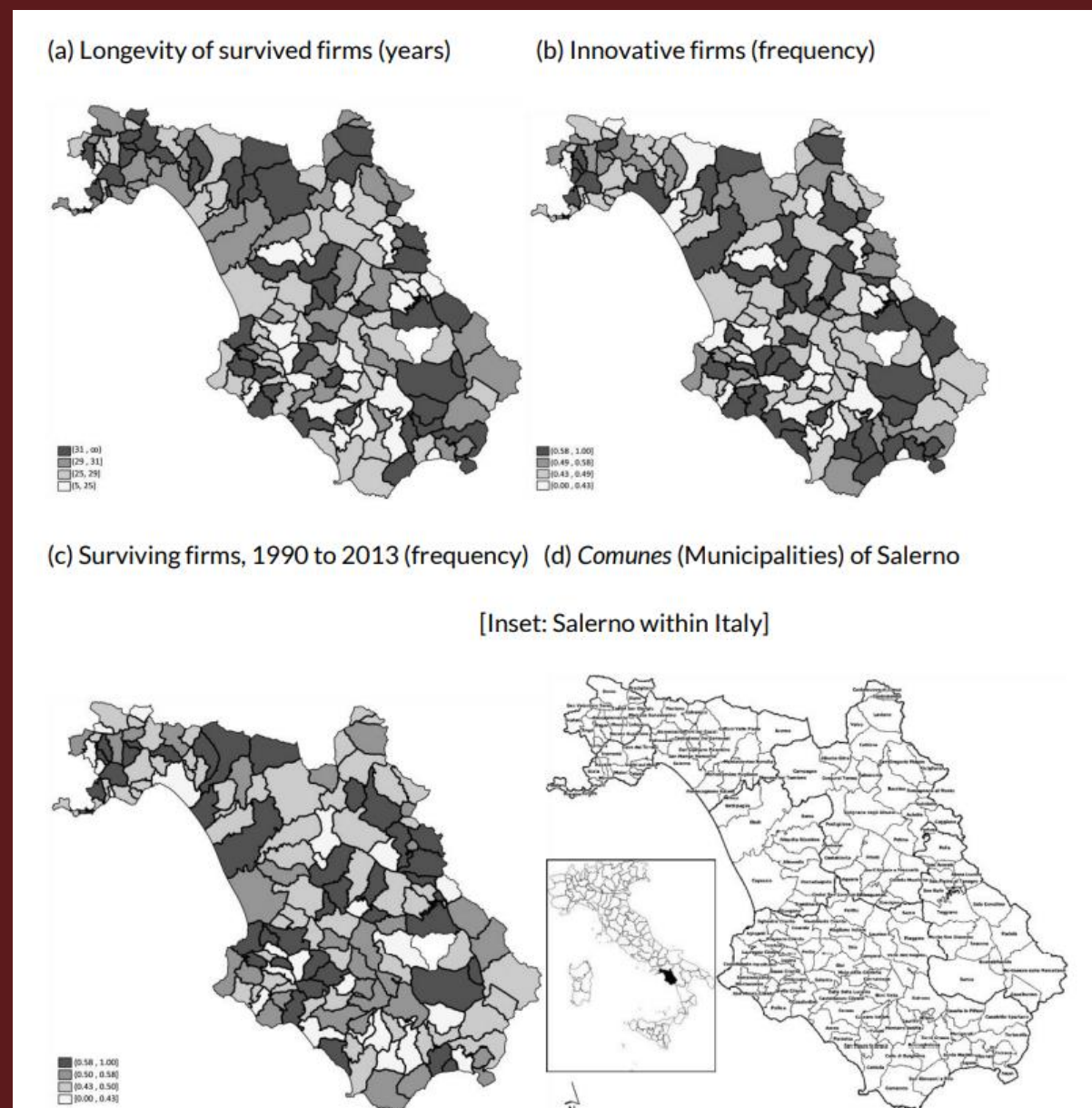
Cottineau, C., Finance, O., Hatna, E., Arcaute, E., & Batty, M. (2016). Defining urban clusters to detect agglomeration economies. *Environment and Planning B: Urban Analytics and City Science*, 46, 1611 - 1626.

- **Research question:** Are agglomeration economies specifically urban? Does the location of economic agents within cities matter? Are larger cities more unequal?
- **Data:** France — 2011 census communes (36546), 2011 tax office household incomes brackets, 2008 CLAP firm-wage data.
- **Method:** log-log scaling regressions, and evaluate across the D-C-P parameter space.
- **Findings:** Wage superlinearity is strongest in dense job centers; population-based effects appear regionally. High incomes cluster in dense cores, and larger cities show a slight but consistent rise in inequality.



SPATIAL AGGLOMERATION, INNOVATION AND FIRM SURVIVAL FOR ITALIAN MANUFACTURING FIRMS (BHATTACHARJEE, MAIETTA & MAZZOTTA, 2023)

- 10 Research Question: How does spatial agglomeration affect firm innovation and survival in Italian manufacturing
- 10 Data: Firm-level panel data from southern Italy
- 10 Spatial econometric (Spatial Durbin) models to measure local spill-over effects
- 10 Findings: Firms near other innovative firms are more likely to innovate and survive
- 10 Conclusion: Local innovation clusters strengthen regional competitiveness and firm resilience



R. Martin & P. Sunley (2023):

Capitalism Divided? London, financialisation and the UK's spatially unbalanced economy

<https://www.tandfonline.com/doi/epdf/10.1080/21582041.2023.2217655?needAccess=true>

1. Research question:

- Has London's economic success benefited the regions?
- Trickle down effect or „finance curse“?
- Has the government's „Levelling up“ policy been successful?

2. Data:

- Historical time series data
- Trade balance statistics
- Regional per capita Gross-value added (GVA)

3. Methodology:

- Historical trend analysis ← time series reaching back as far as the 1870s
- Literature analysis ← integrating other author's finding regarding the „finance curse“
- Policy critique

4. Findings:

- London's lead has been increasing since the 80s (25% of the UK's GDP and 38% of Englands GDP)
- Misallocation of capital (London receives highest per capita amount of gov. spending)
- Insufficient policies, deeply institutionalised problems
- Gravitational pull

„At the same time, regional and sub-regional disparities in economic prosperity in the UK are amongst the largest in the OECD nations, far greater than in other major advanced economies such as Germany, France, Italy, Japan and the USA, and only exceeded by those in Hungary, Turkey and Colombia (OECD, 2020).“

Summary

The study investigates the extent to which **Smart Specialisation Strategies** (S3, EU policy for regional innovation) **align with the actual technological and other relevant capacities of European regions.**

Data:

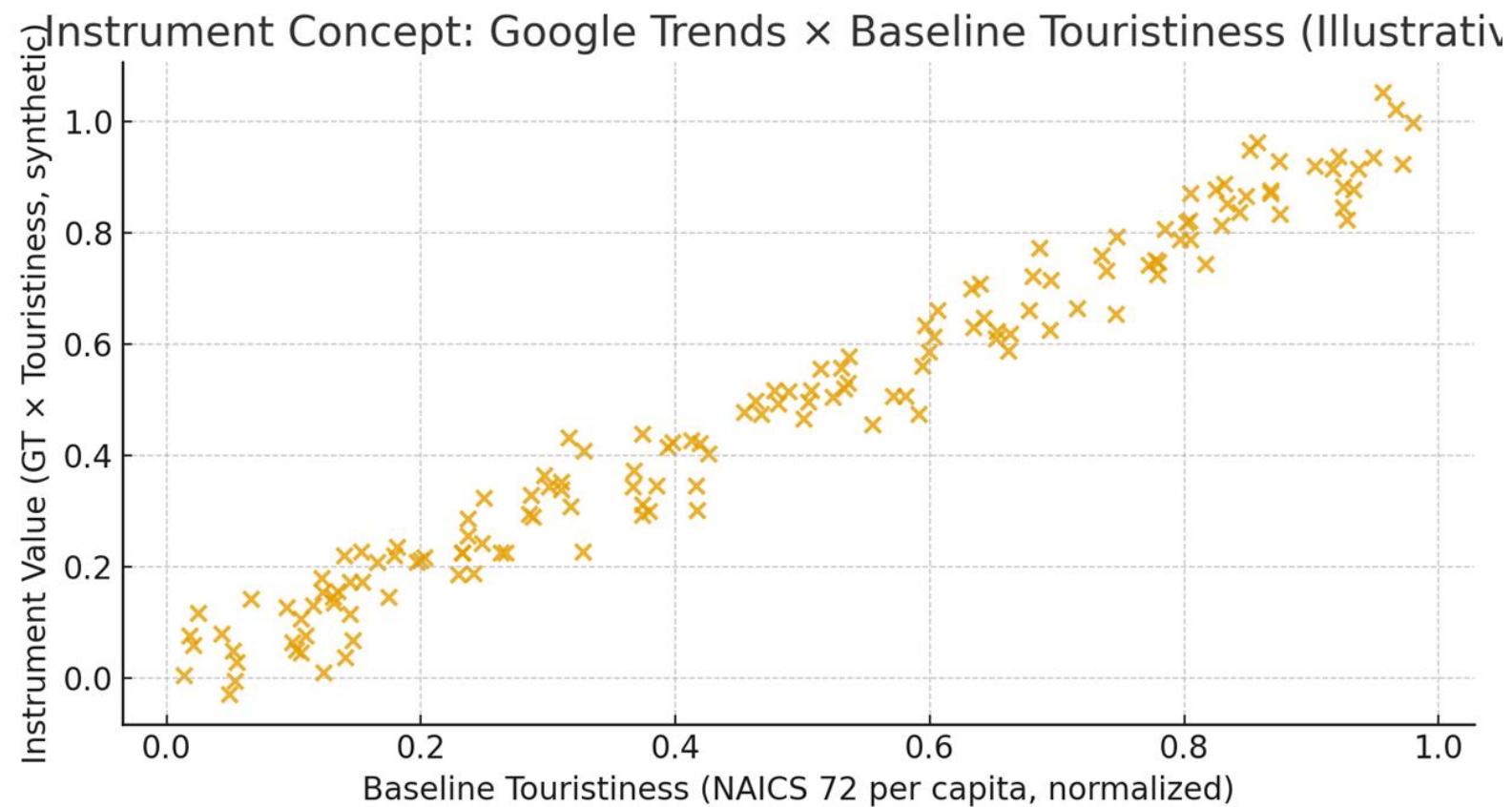
- 164 European regions, scrutinising the period from 2011-2015.
- European Patent Office PATSTAT database – **technological knowledge production via patents**
- European Commission S3 platform – **S3 priority sectors**

Methodology: evolutionary economic geography and network analysis

- Framework 1: **Centrality Analysis** – tests whether S3 policies focus on central technologies of the local innovation network.
- Framework 2: **Specialisation and potential analysis** – assesses whether S3 priorities focus on existing strengths or potential diversification opportunities.

The study found that **many S3 strategies do not align with the regional capabilities. Innovation remains spatially concentrated** in regions with dense and diverse knowledge places, with **institutional inequality** having a strong influence on that persistence.

Data Visual 2 — ZRI, ZHVI, and Price-to-Rent (Illustrative)



COVID-19 Spread in Beijing

Research paper: [Spatial distribution characteristics of the COVID-19 pandemic in Beijing and its relationship with environmental factors – Yi Han et al. 2021](#)

Research question:

- How did COVID-19 spread in Beijing, and what environmental factors affected this spread?

Data Used:

- COVID-19 case numbers in Beijing from June 11 to July 5, 2020.
- Environmental factors like population density, distance to Xinfadi market, and other city features.

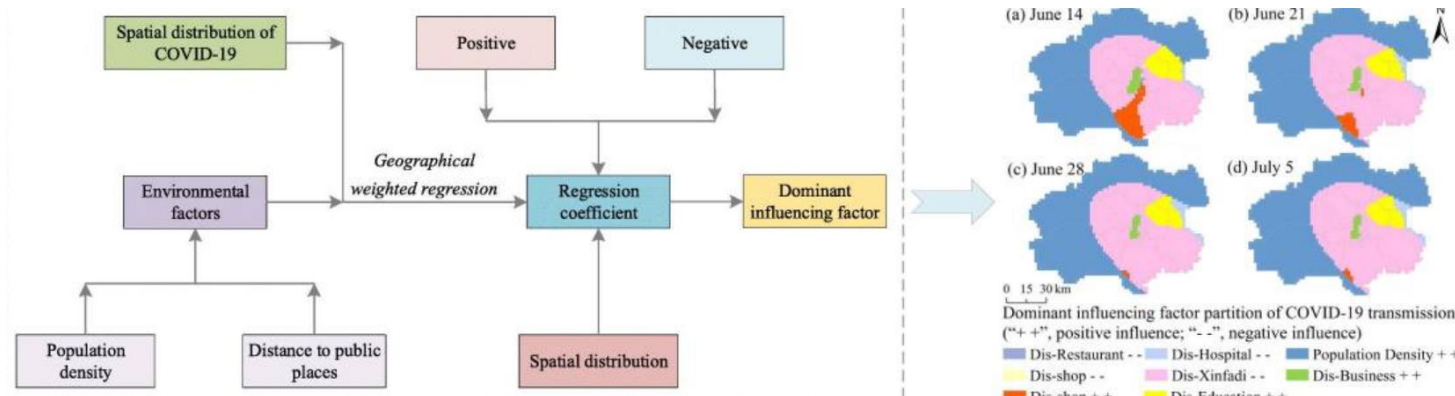
Methods:

- Checked for clusters using spatial autocorrelation.
- Used Spearman correlation to see which factors were related to case numbers.
- Applied Geographically Weighted Regression (GWR) to see how factors influenced different areas.

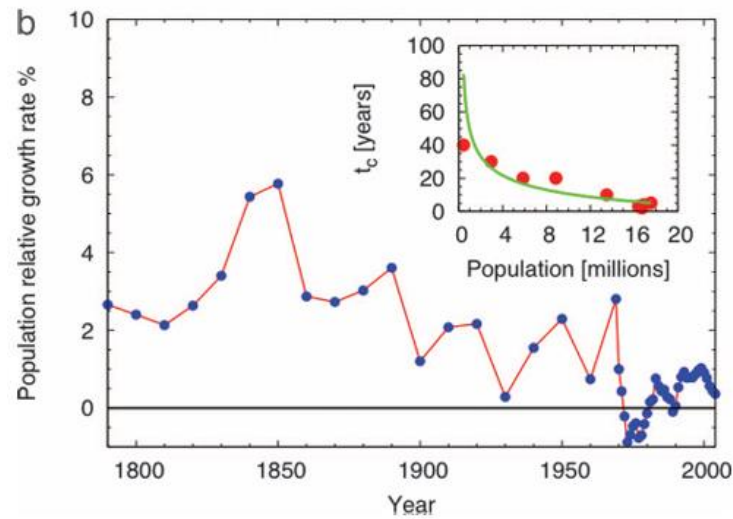
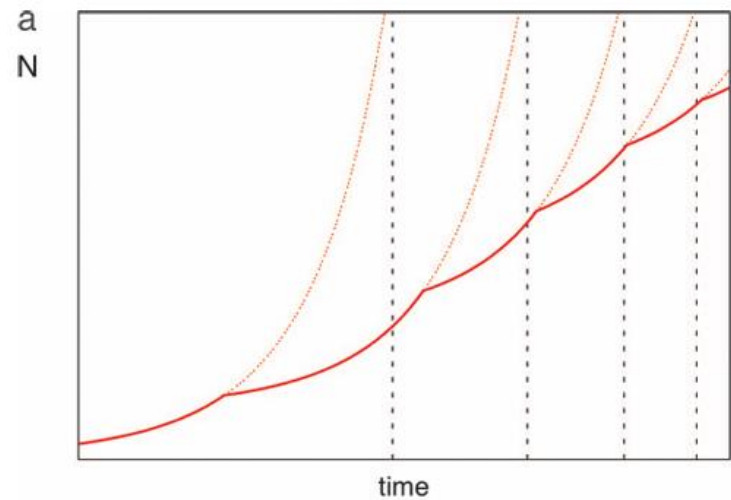
Findings:

- COVID-19 spread in two main phases: first spreading outward, then concentrating near existing cases.
- Areas around the Xinfadi market had higher case clusters, showing it was a hotspot.
- Population density and closeness to the market were the strongest factors affecting the spread.

Graphical abstract:



Innovation cycles to sustain superlinear cities

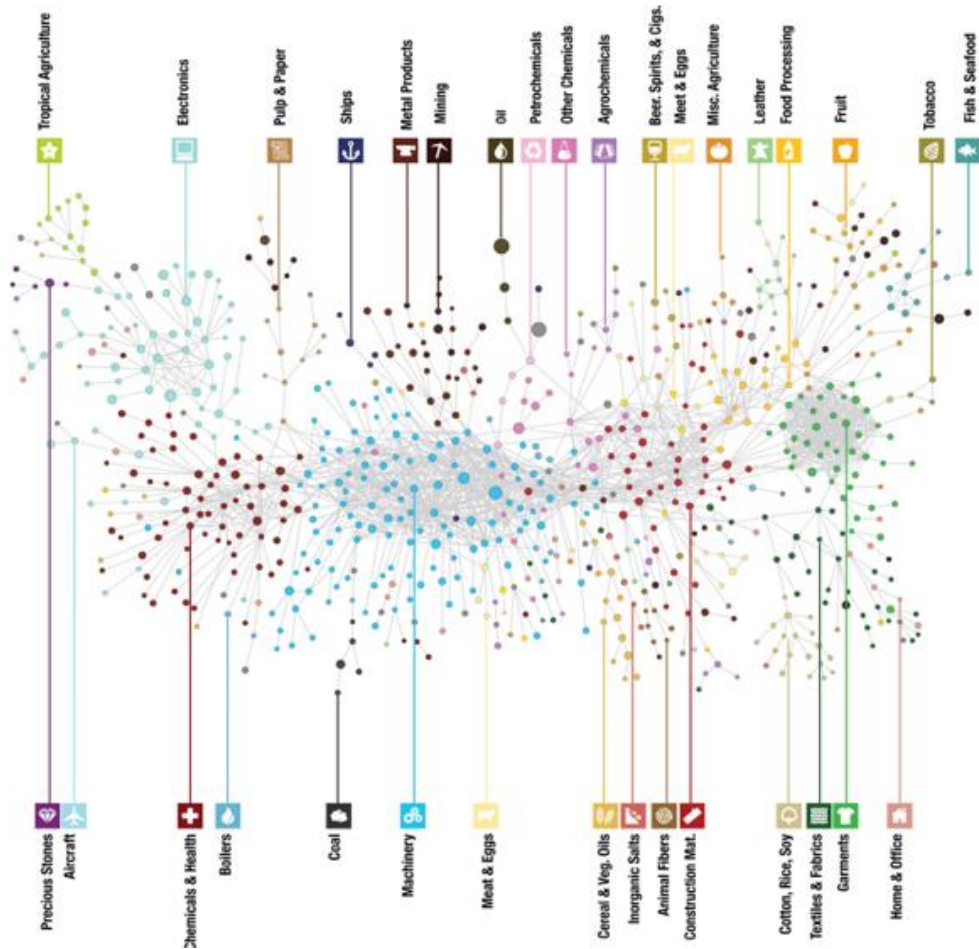


Today

1. Agents of change
2. Collaboration networks in diversification
3. Life cycles

1. Agents of change

How do we jump big in the product/technology/knowledge space?

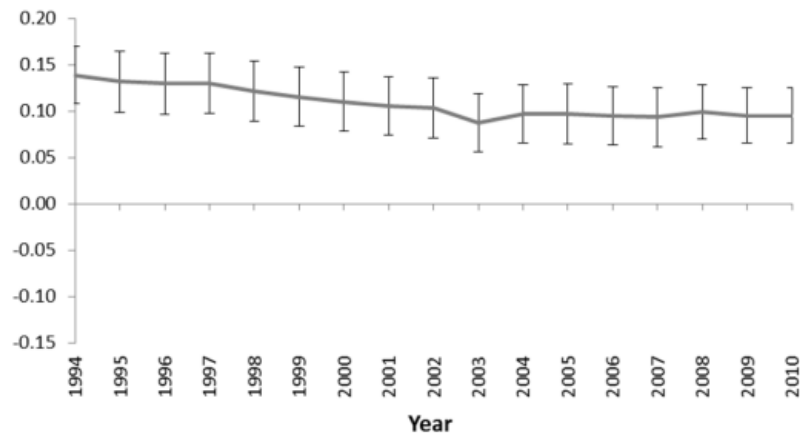


Unrelated diversification

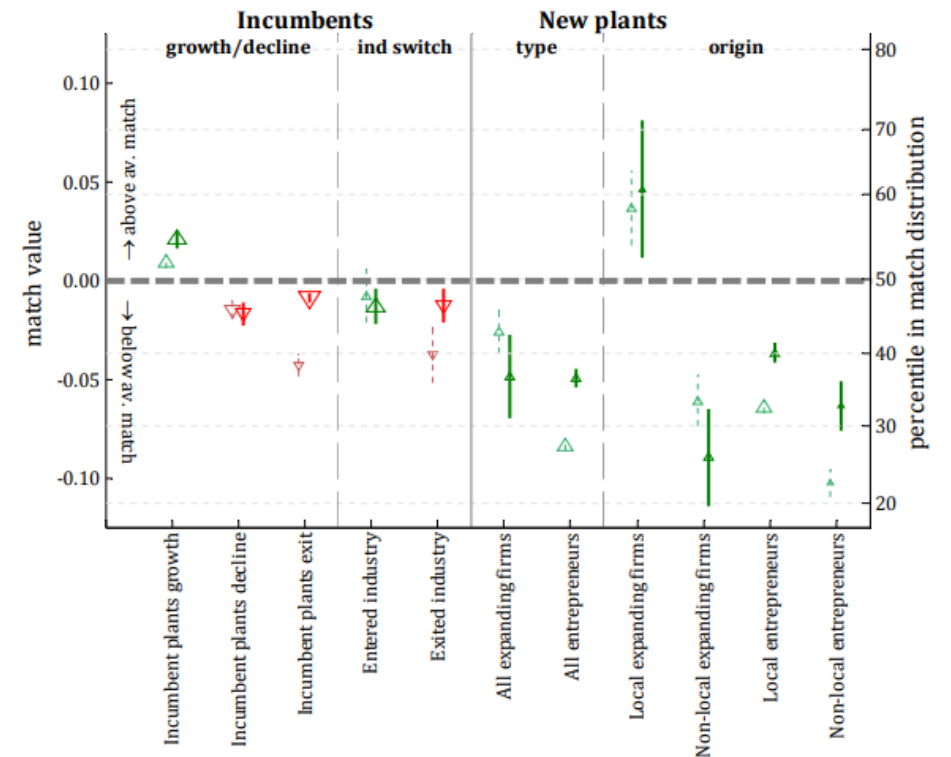
- Investments?
 - FDI
 - Governments
- Migration?
 - Inventors
 - Scholars
 - Businessmen

Structural change by type of firms

Figure 5: Structural change in Sweden's labor market regions

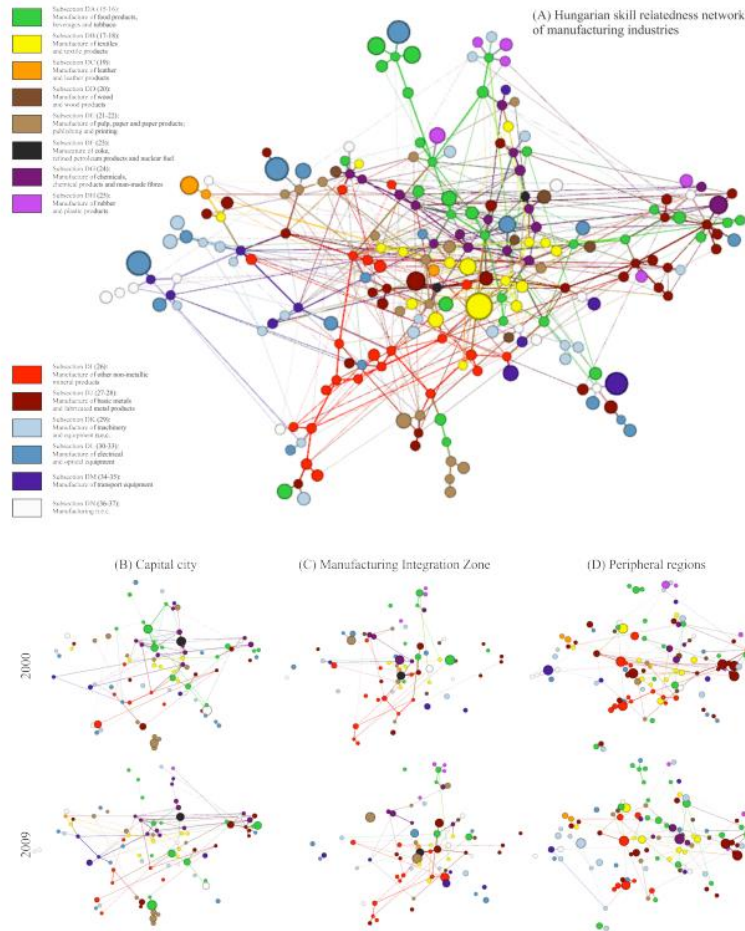


The graph depicts the development of the average capability match of a regions' local industries to the local economy of the base year 1994 ($S_{rt,1994} = \sum_i \frac{E_{it}}{E_{rt}} \bar{Q}_{ir,1994}^{rel}$), including 95% error bars. In the presence of structural change, $S_{rt,1994}$ should fall (diversification) or rise (increasing of existing focus) over time. The error bars depict a 95% confidence interval calculated as ± 1.96 times the standard deviation of the mean structural change across regions



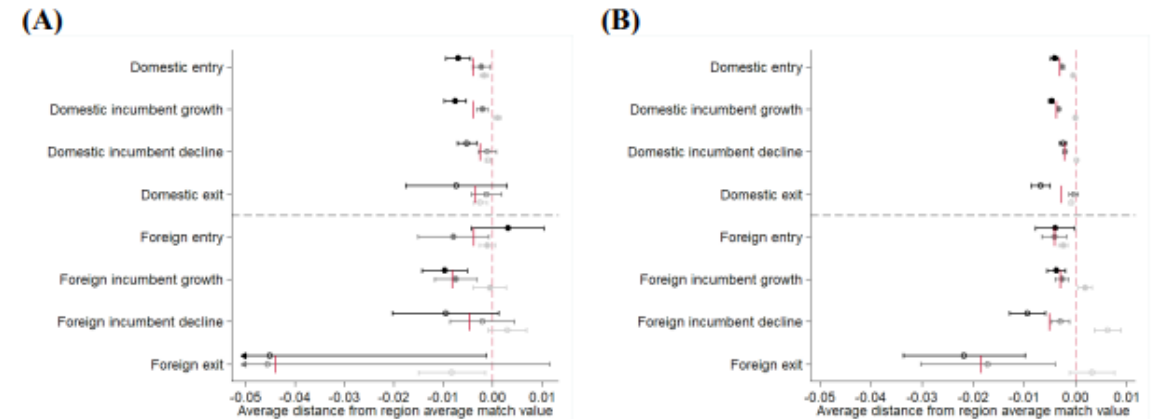
The markers shows for each agent type the employment weighted average capability match of the local industries in which the agent type creates or destroys employment within ten years after being founded, averaged across all establishments of the agent minus the region's coherence. That is, the graph shows by how much an agent's average capability match exceeds the region's coherence. As a consequence, employment created (destroyed) at $\bar{A} > 0$ corresponds to a diversification (further focusing) of the regional capability base. Employment-creating agents are denoted with a green, upward pointing arrow, employment-destroying agents with a red downward-pointing arrow. The error bars depict a 95% confidence interval, based on the standard deviation of the mean capability match across all establishments of a given agent type. To facilitate interpretation of $\bar{A}$ -values, the right most vertical axis provides the corresponding percentiles of a capability match value in the overall distribution of the 1994 economy. The one-year figures reported in Figure 6 are provided as a reference with lighter markers and dotted confidence intervals.

Foreign firms and structural change



Notes: (A) The network representation was constructed by superimposing the strongest 1.5% of edges on the maximum spanning tree of the skill relatedness network. ForceAtlas2 layout in Gephi software was used as layout. After obtaining the network layout, additional ties up to the strongest 5% of all edges were added to the graph. Node size is proportional to the number of employees in each industry. (B-D) Node size represents employment location quotient; only those industries are represented that have $LQ > 1$.

Figure 6. Structural change by agent type in the 3 region groups



Notes: (A) Short-term (1-year) structural change. (B) Long-term (10-year) structural change. The red dashed line indicates the average distance of agent types from the region average match value. Values to the right of this line indicate more related (*i.e.* above average) diversification, while values to the left indicate more unrelated (*i.e.* below average) diversification. Error bars indicate 95% confidence intervals. The base year is 2000. Black error bars depict the *Manufacturing Integration Zone*, dark grey error bars indicate the *peripheral regions*, light grey bars signify the *capital city*. Red solid vertical lines indicate the agent type-average structural change.

2. Networks

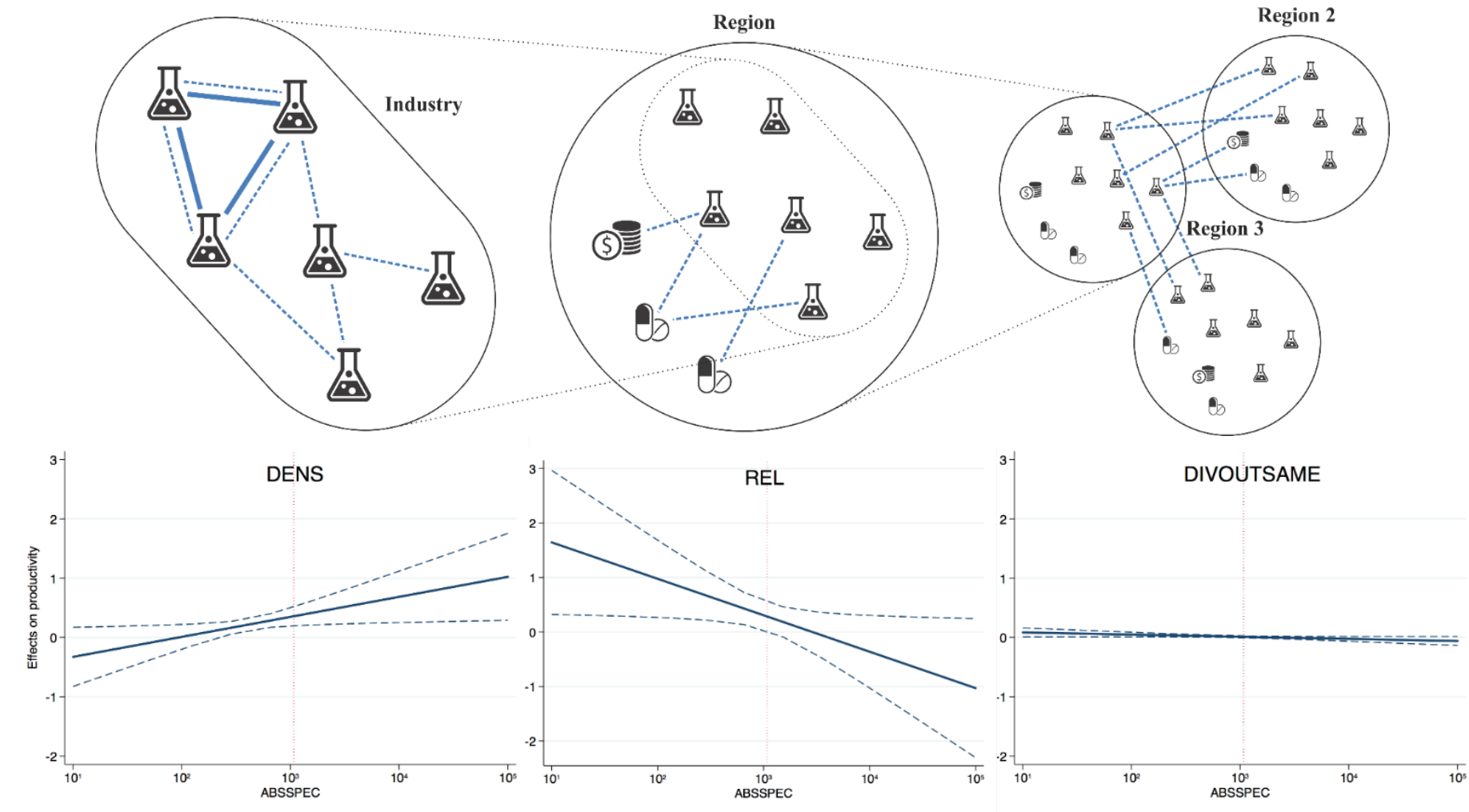


Local capabilities and interregional connections



Knowledge sourcing in diversification

- The effect of co-worker networks on regional growth of income depends on region size
- (ABSSPEC)

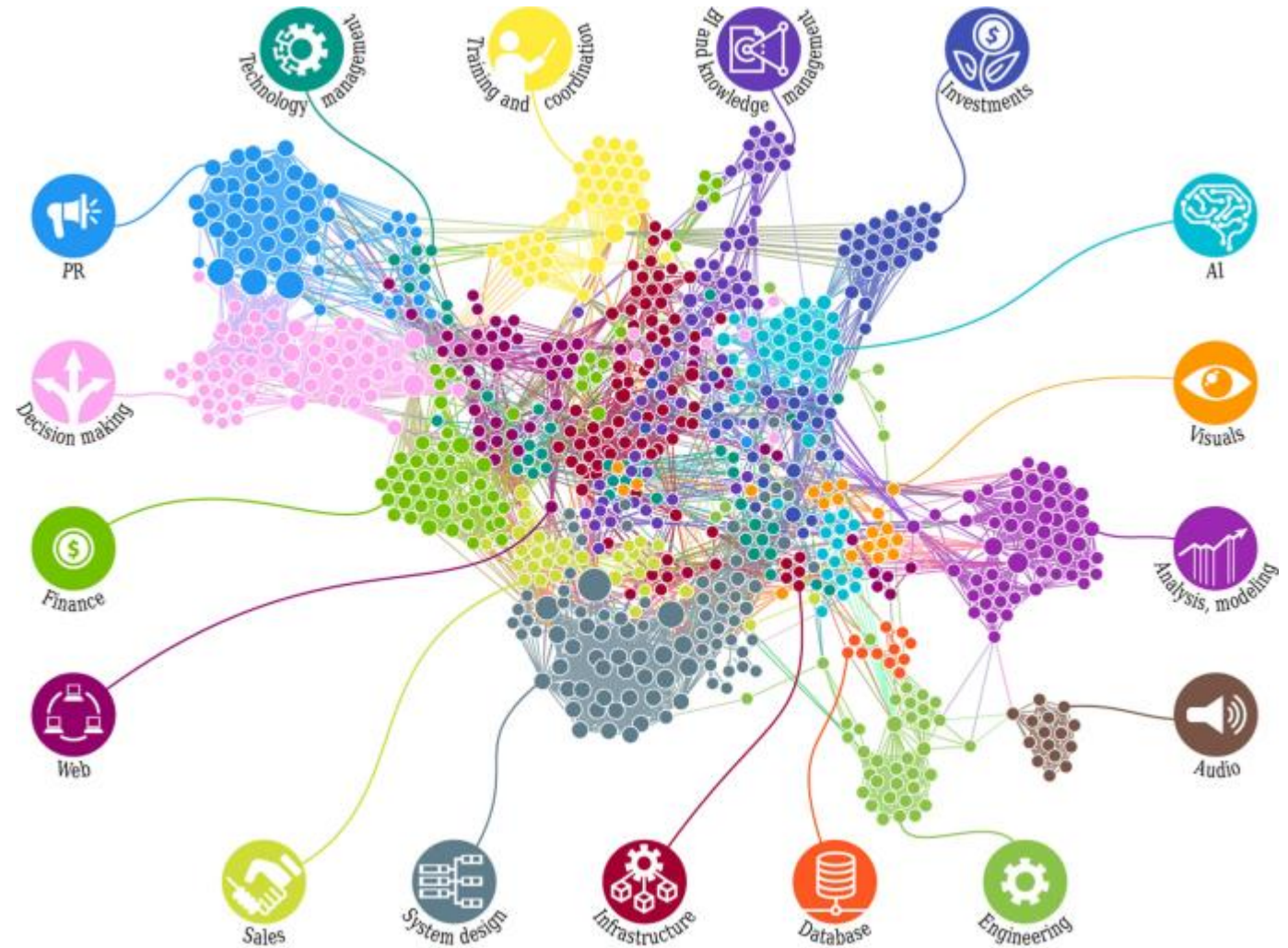


Question 1

**Can interregional
networks**

substitute

**local capabilities in
diversification?**



Firms as drivers of diversification

Agents of structural change: mobile entrepreneurs (Neffke et al., 2018); foreign-owned firms (Elekes et al., 2019), R&D investments (Crescenzi et al., 2022).

Interregional knowledge sourcing dominantly happens within boundaries of multilocation companies (Frigon and Rigby, 2022; Zhang and Rigby, 2022).

Multilocation companies intensify collaboration to source from diversity (Alcacer and Zhao, 2012).

Firms as drivers of diversification

Exploitation of the local knowledge base favors related diversification (Wang and Zhao, 2018).

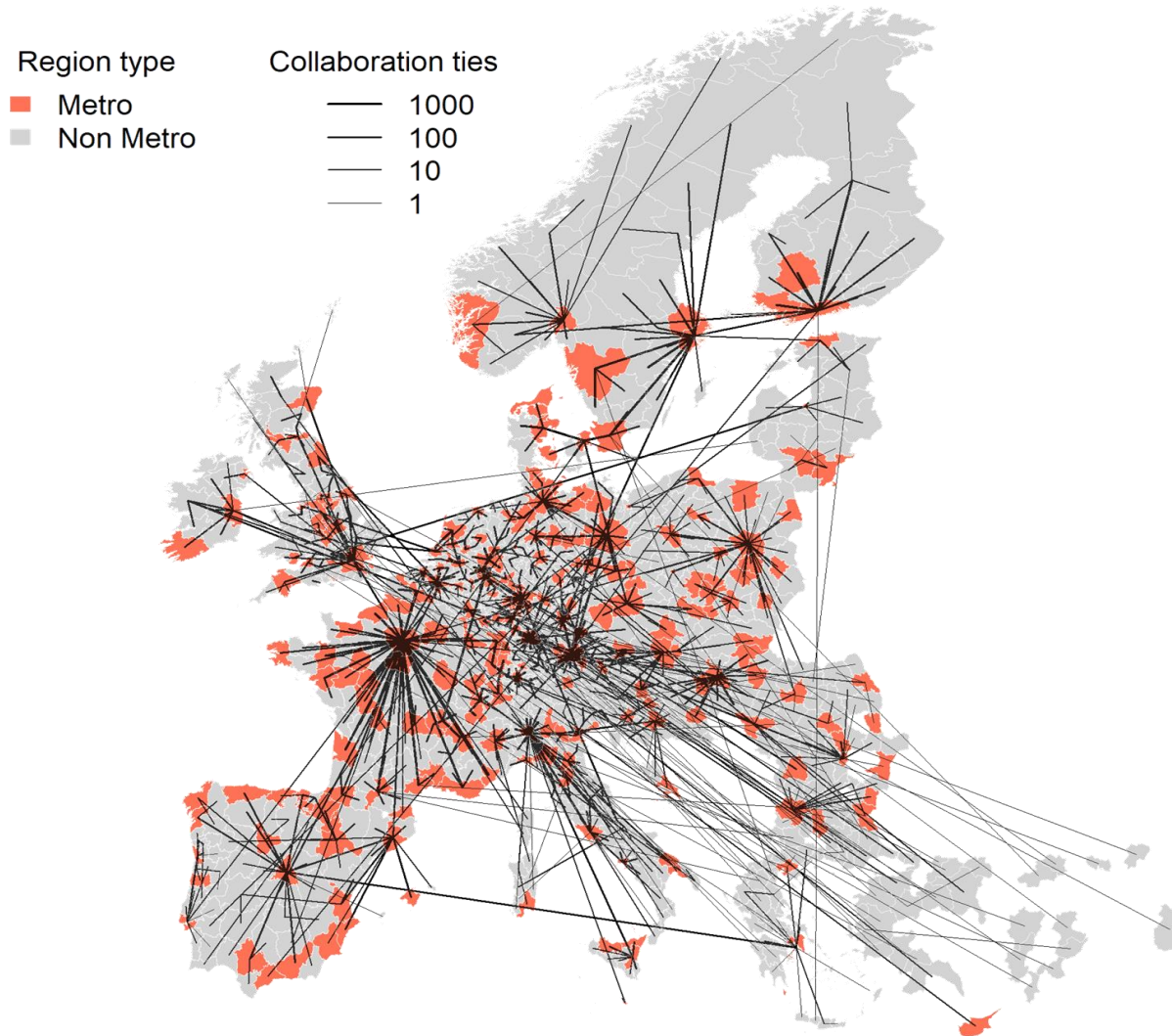
Agents of structural change and co-worker networks literature suggest that regions can source unrelated knowledge from the diversity of regions that multilocation companies connect (Neffke et al., 2018; Elekes et al., 2019; Eriksson and Lengyel, 2019; Crescenzi et al., 2022).

Question 2

**Do internal connections
of multilocation firms across
regions
favors diversification
into related or unrelated new
activities?**



Data



- EPO PATSTAT
- 1981-2015, five-year periods
- 4-digit CPC codes
- NUTS3 regions (EUROSTAT classificaion of metro/nonmetro)
- Fractionally split patents to regions by inventor addresses

Probability of ENTRY

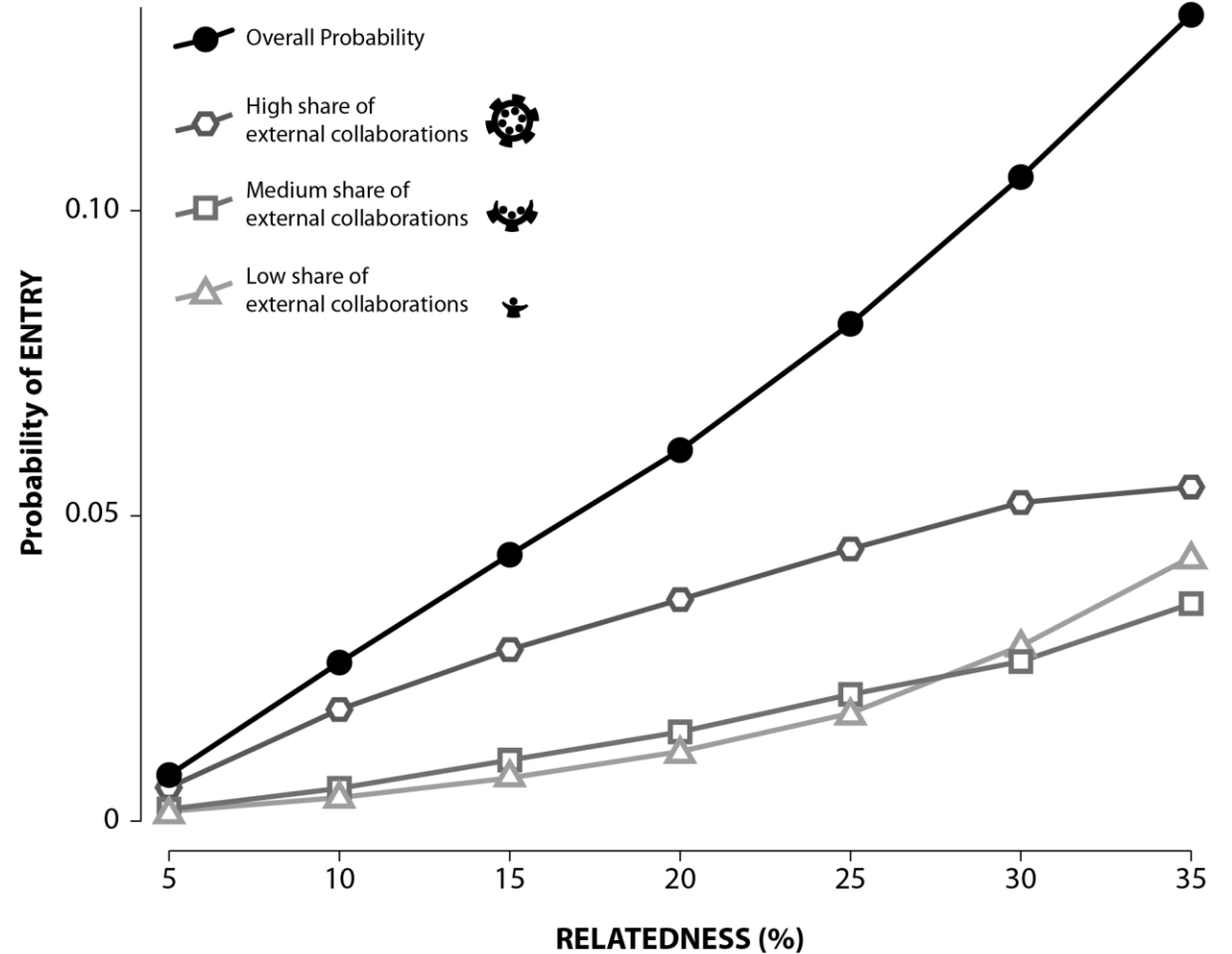
$$ENTRY_{i,r,t} = \begin{cases} 1 & \text{if } RTA_{i,r,t-1} < 1 \text{ and } RTA_{i,r,t} > 1 \\ 0 & \text{otherwise} \end{cases}$$

$$RTA_{i,r,t} = \frac{\text{patents}_{r,t} \frac{i}{\sum_i \text{patents}_{r,t}} (i)}{\sum_r \text{patents}_{r,t} \frac{i}{\sum_r \sum_i \text{patents}_{r,t}} (i)}$$

$$RELATEDNESS_{i,r,t} = \frac{\sum_{j \in r} x_j \varphi_{ij,t}}{\sum_j \varphi_{ij,t}} \times 100$$

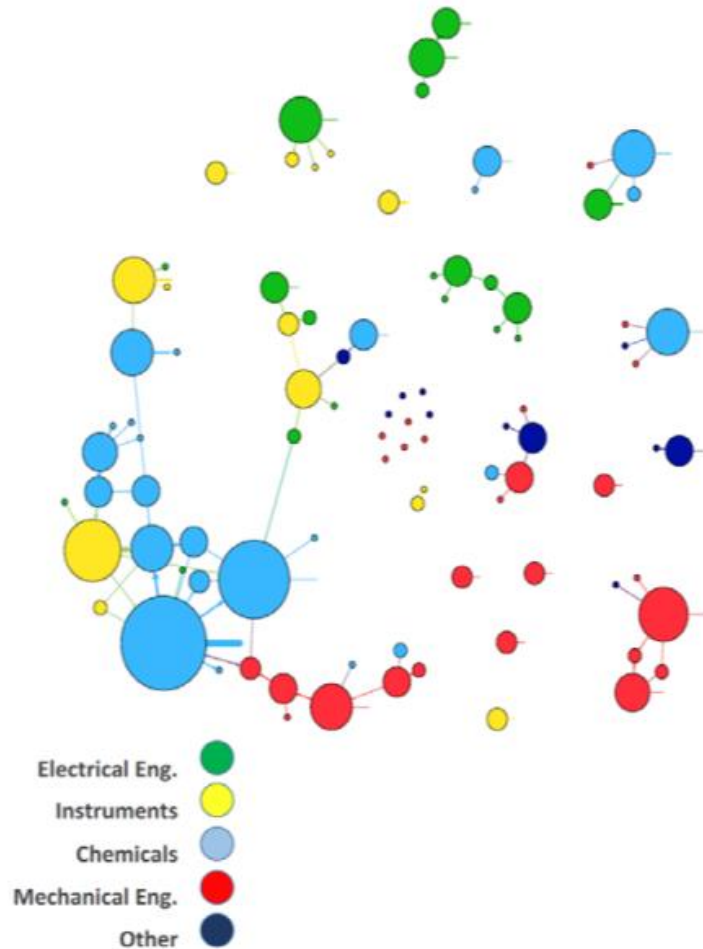
$$\varphi_{ij,t} = \min\{P, (R|TAx_{i,t}RTAx_{j,t}) \{P(R|TAx_{j,t}RTAx_{i,t})\}\}$$

$$EXTCOLLAB_{i,r,t} = \frac{EL_{i,r,t}}{EL_{i,r,t} + IL_{i,r,t}}$$

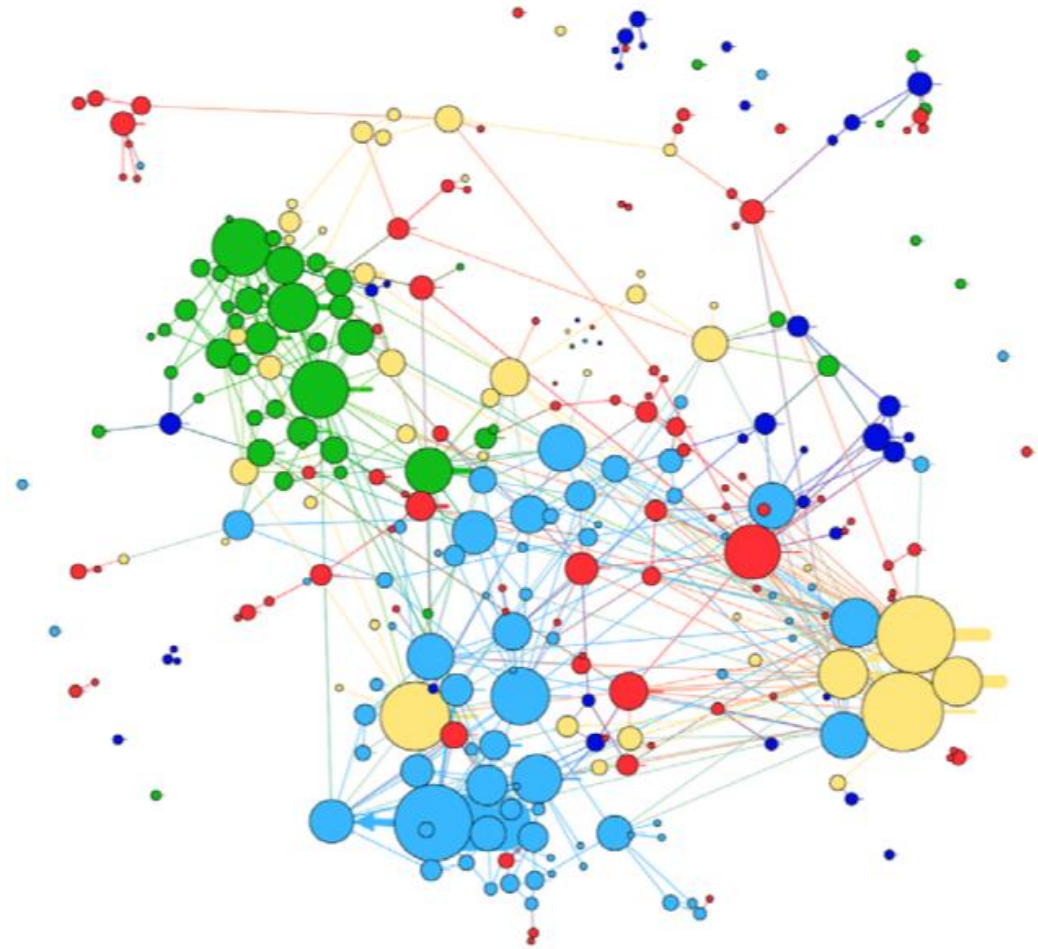


The knowledge space

1981 – 1985



2001–2005



Regression analysis

Regress the probability of ENTRY on NETWORK variables and controllers.

- Region-technology fixed effects (time fixed effects).
- LPM fit (logit).
- Lagged independent variables.

$$P(ENTRY_{i,r,t} = 1) = \alpha + \beta X_{i,r,t-1} + \gamma Z_{i,r,t-1} + \mu_{i,r,t} + \varepsilon_{i,r,t}$$

$$DIVERSITY_{i,r,t} = \frac{-\sum l_{rq} \times \log(l_{rq})}{\log(k)}$$

$$INTENSITY_{i,r,t} = \frac{\sum_q L_{f \in i, r=f \in q}}{\sum_{q, f \in i, r=f \in q} N_{f \in i, r} \times N_{f \in q}}$$

Explanatory and control variables

Variable Name	Description
RELATEDNESS	Relatedness density (Hidalgo et al. 2007) calculated on the knowledge-space network. Measure of how close an emerging technology i is to other existing technologies in the region at time t .
EMPLOYMENT	Total number of employees in a region.
FIRMS	The number of firms in a region and CPC class that have been assigned at least one patent until t .
NEWFIRMS	The number of firms in a region and CPC class that have been assigned a first patent since $t-1$.
MULTILOC	The share of firms in the region and CPC class that employ inventors in other regions as well.
INVPERFIRM	The number of inventors per number of firms that are assignees of patents in the region and CPC class at period t .
EXTCOLLAB	The ratio of interregional collaborations among all collaborations of the inventors living in the region and patenting in a CPC class in period t .
DENSITY	The ratio of observed co-inventor collaboration among all possible collaborations the inventors in a specific CPC class could have within the region.
DIVERSITY	Entropy of the aggregated co-inventor connections from a CPC class and the region to other regions at period t .
INTENSITY	The ratio of observed links within the firm to other regions among all possible such links. This measure is calculated at the CPC and region level at period t .

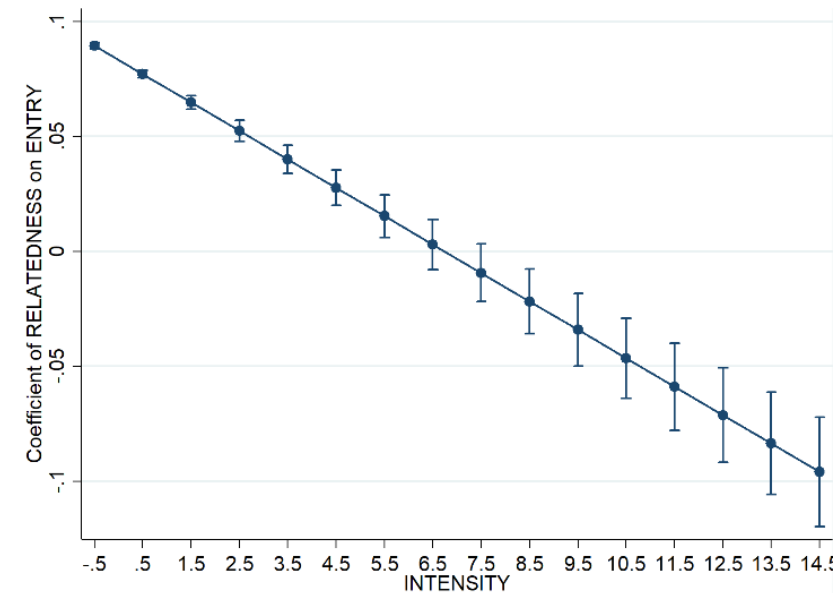
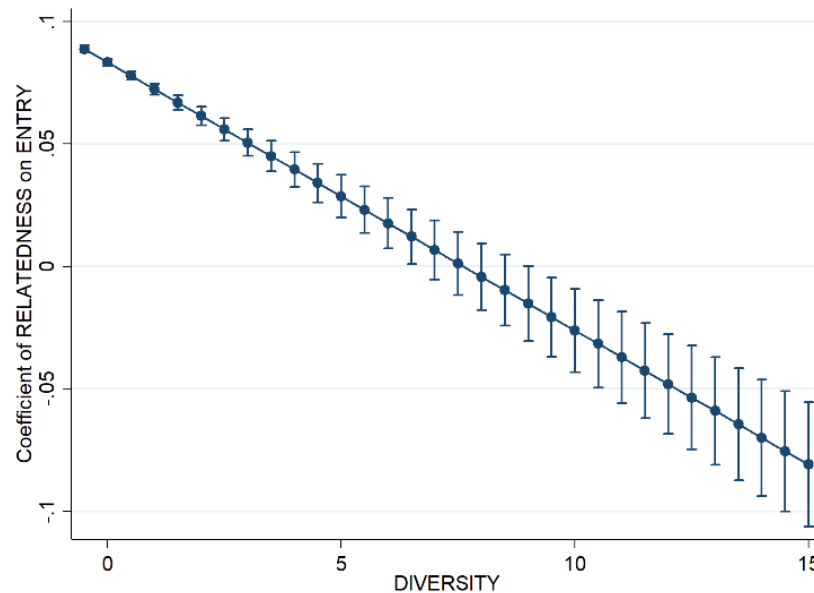
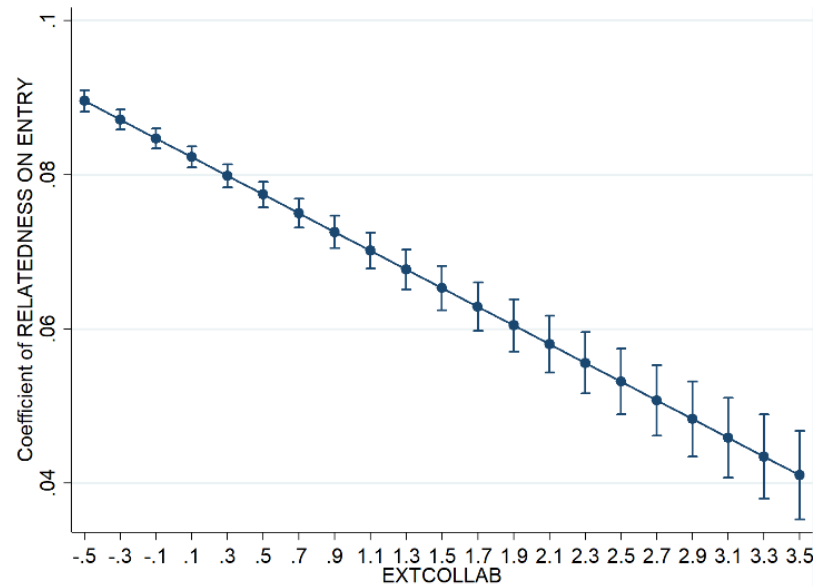
	(1) LPM	(2) LPM	(3) LPM	(4) LPM	(5) LPM	(6) LPM	(7) LPM	(8) LPM	(9) LOGIT	(10) LOGIT
	ALL REGIONS					METRO	NONMETRO	ALL REGIONS		
RELATEDNESS	0.084*** (0.001)	0.084*** (0.001)	0.087*** (0.001)	0.084*** (0.001)	0.087*** (0.001)	0.084*** (0.001)	0.089*** (0.001)	0.072*** (0.001)	0.726*** (0.006)	0.541*** (0.010)
EMPLOYMENT	0.025*** (0.001)	0.025*** (0.001)	0.024*** (0.001)	0.025*** (0.000)	0.024*** (0.001)	0.033*** (0.001)	0.016*** (0.001)	0.000 (0.001)	0.834*** (0.022)	0.350*** (0.021)
INVPERFIRM	-0.007** (0.002)	-0.005* (0.002)	-0.003 (0.002)	-0.007** (0.002)	-0.005* (0.002)	-0.006* (0.003)	-0.003 (0.004)	-0.003 (0.002)	-0.004 (0.009)	0.015 (0.009)
FIRMS	-0.023*** (0.002)	-0.023*** (0.002)	-0.013*** (0.002)	-0.024*** (0.002)	-0.014*** (0.002)	-0.012*** (0.003)	-0.016*** (0.004)	-0.009*** (0.002)	-0.082*** (0.011)	-0.054*** (0.011)
NEWFIRMS	0.014*** (0.003)	0.014*** (0.003)	0.010** (0.003)	0.015*** (0.003)	0.006 (0.003)	0.005 (0.004)	0.004 (0.007)	0.002 (0.003)	0.065*** (0.014)	0.015 (0.014)
MULTILOC	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.004*** (0.001)	0.003*** (0.001)	0.003*** (0.000)	0.002 (0.002)	-0.004 (0.003)
DENSITY	0.003** (0.001)	0.003** (0.001)	0.003** (0.001)	0.003** (0.001)	0.003** (0.001)	0.003* (0.001)	0.002 (0.002)	0.001 (0.001)	-0.008* (0.004)	-0.022*** (0.004)

	(1) LPM	(2) LPM	(3) LPM	(4) LPM	(5) LPM	(6) LPM	(7) LPM	(8) LPM	(9) LOGIT	(10) LOGIT
	ALL REGIONS					METRO	NONMETRO		ALL REGIONS	
RELATEDNESS	0.084*** (0.001)	0.084*** (0.001)	0.087*** (0.001)	0.084*** (0.001)	0.087*** (0.001)	0.084*** (0.001)	0.089*** (0.001)	0.072*** (0.001)	0.726*** (0.006)	0.541*** (0.010)
EXTCOLLAB		0.002* (0.001)	0.030*** (0.002)							
RELATEDNESS × EXTCOLLAB			-0.029*** (0.002)							
DIVERSITY				0.001 (0.001)	0.021*** (0.003)	0.023*** (0.004)	0.002 (0.007)	0.018*** (0.003)	0.153*** (0.014)	0.038** (0.014)
INTENSITY				0.001 (0.001)	0.023*** (0.003)	0.030*** (0.004)	0.021*** (0.005)	0.005 (0.003)	0.208*** (0.012)	0.017 (0.012)
RELATEDNESS × DIVERSITY					-0.018*** (0.003)	-0.019*** (0.003)	0.005 (0.008)	-0.009** (0.003)	-0.178*** (0.013)	-0.072*** (0.012)
RELATEDNESS × INTENSITY					-0.023*** (0.002)	-0.027*** (0.003)	-0.024*** (0.004)	-0.006* (0.003)	-0.230*** (0.011)	-0.049*** (0.011)
CONSTANT	0.134*** (0.000)	0.135*** (0.000)	0.136*** (0.000)	0.134*** (0.000)	0.136*** (0.000)	0.122*** (0.001)	0.135*** (0.001)	0.228*** (0.009)		
Period FE	NO	NO	NO	NO	NO	NO	NO	YES	NO	YES
N	2,418,740	2,418,740	2,418,740	2,418,740	2,418,740	613,604	1,805,136	2,418,740	762,490	762,490
R-sq	0.017	0.017	0.017	0.017	0.017	0.016	0.018	0.022		
AIC	150,348.7	150,329.5	149,423.6	150,348.5	149,314.0	254,731.8	-169,679.4	138,328.0	520,371.7	511,581.8
ll	-75,167.3	-75,156.8	-74,702.8	-75,165.2	-74,659.5	-127,354.9	84,850.7	-69,148.0	-260,174.9	-255,774.9

Interregional collaboration as substitution for relatedness?

Coefficients of RELATEDNESS at levels of EXTCOLLAB, DIVERSITY, INTENSITY.

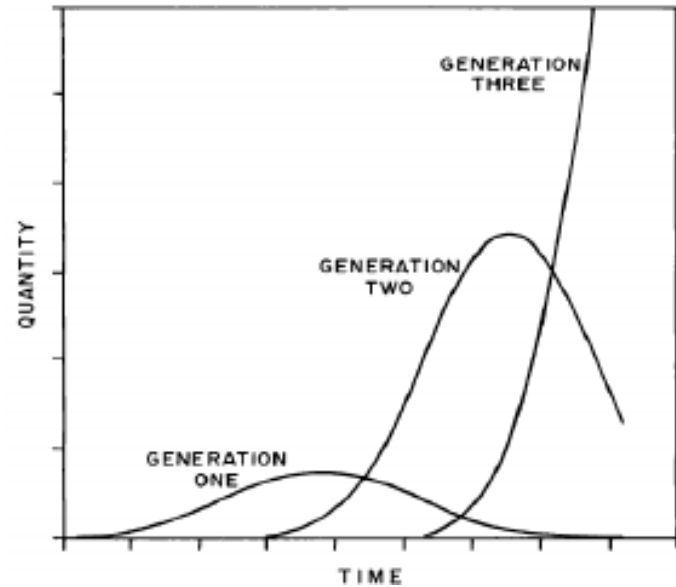
- EXTCOLLAB: RELATEDNESS becomes smaller but remains positive.
- DIVERSITY, INTENSITY: RELATEDNESS becomes negative at their high levels.



3. Spatial life cycles

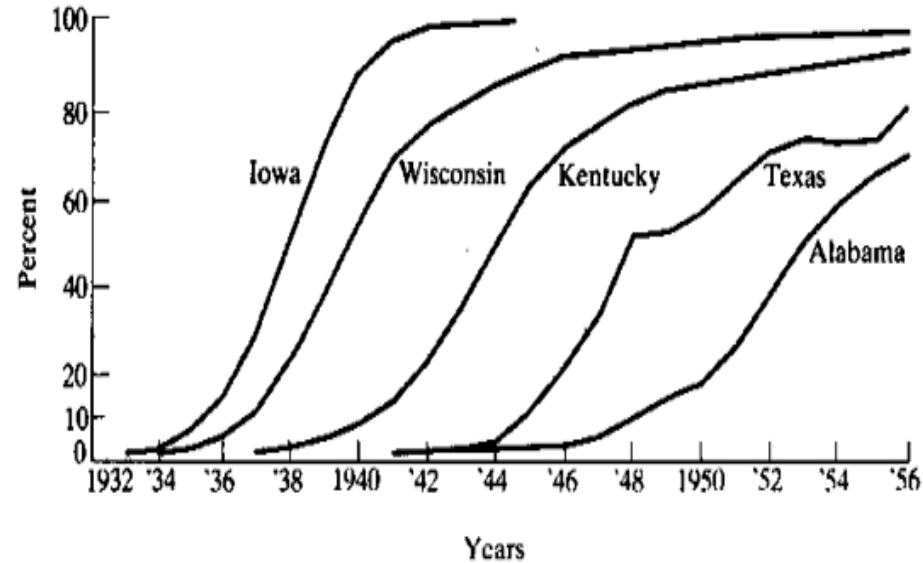
Waves of technology life-cycle

New technologies outcompete old ones.



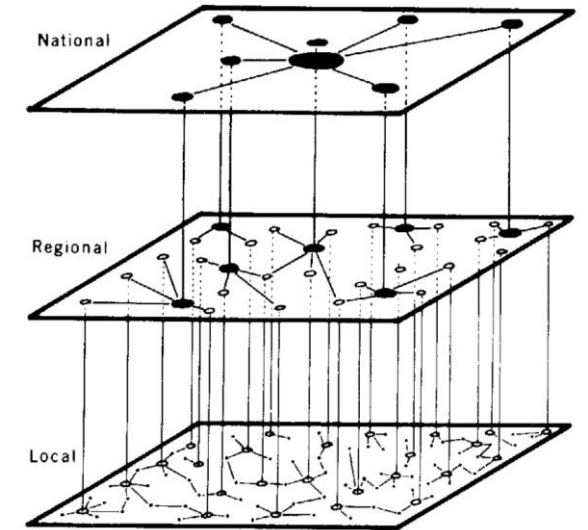
Norton & Bass (1987, Man Sci)

Innovation gets quicker to proximate places.

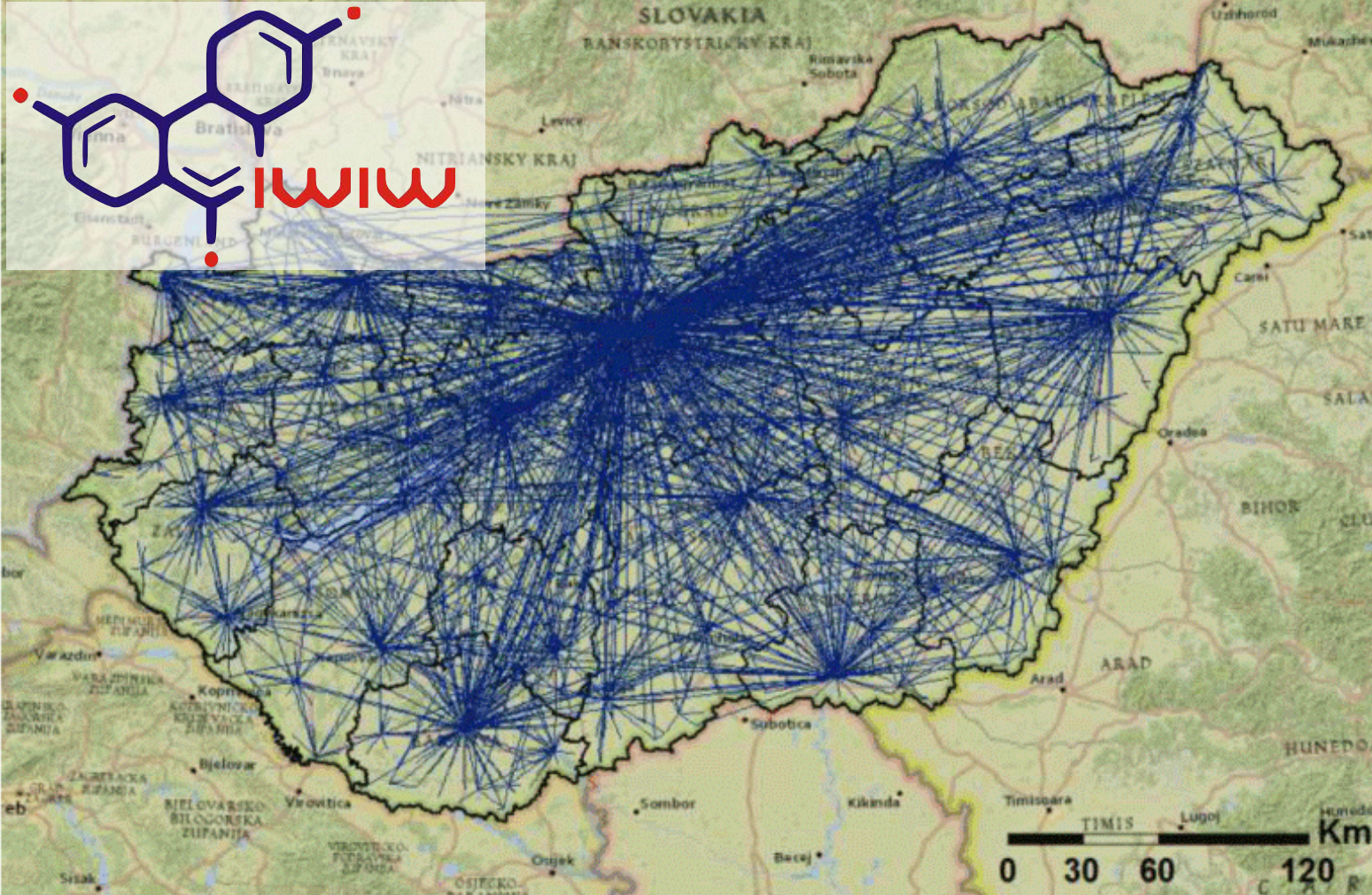


Griliches (1957, Econometrica)

Innovation spreads from big to medium cities and then to small towns.



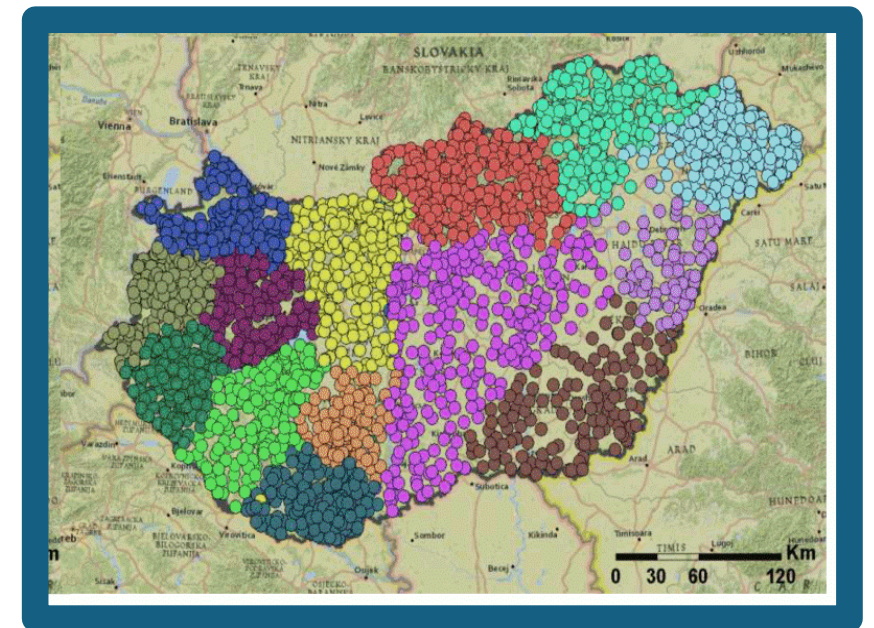
Haegerstrand (1953, Chicago UP)



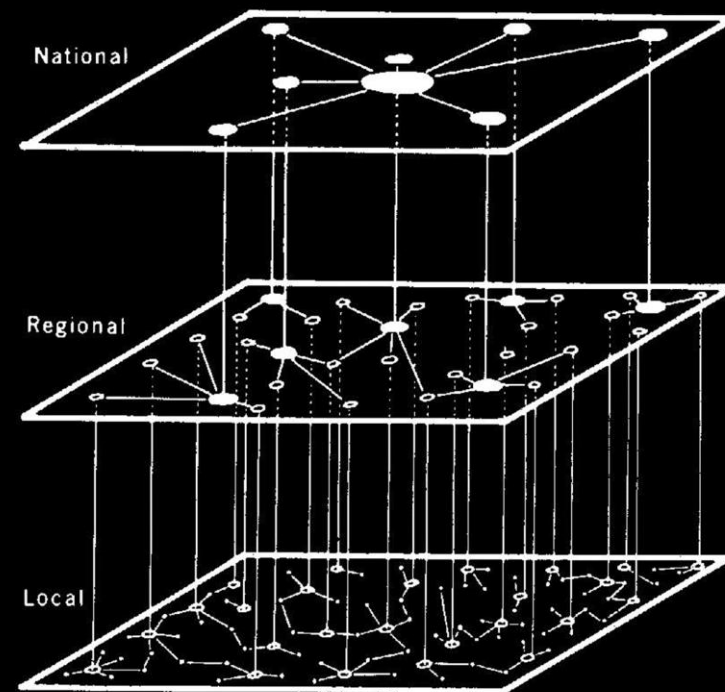
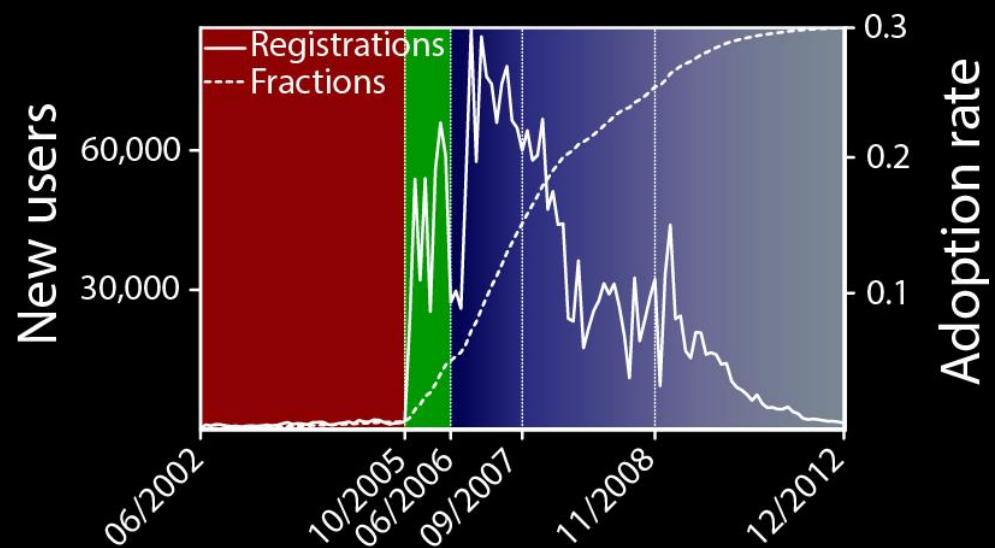
Social media
platform

2.8 M Users

- **Location** (self-reported),
- **Date of registration,**
- **Date of last login,**
- **ID of friends,**
- **ID of invitor**



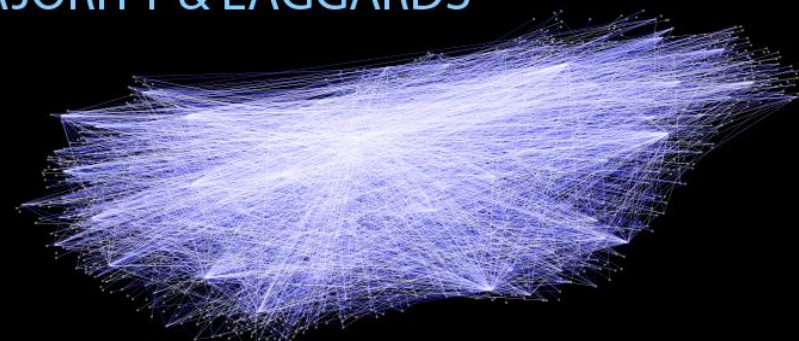
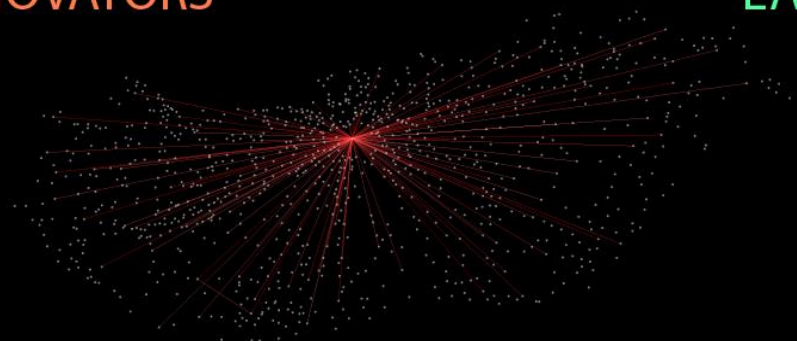
Jakobi and Lengyel (2014) Springer; Lengyel et al. (2015) PLoS ONE



INNOVATORS

EARLY ADOPTERS

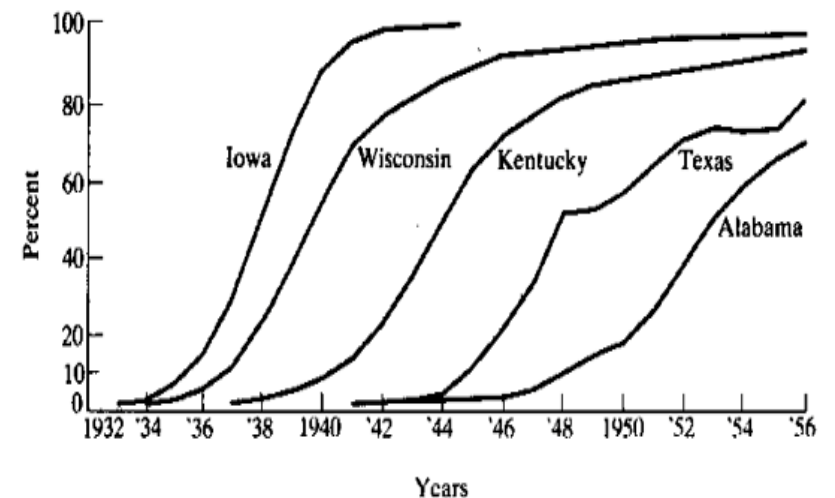
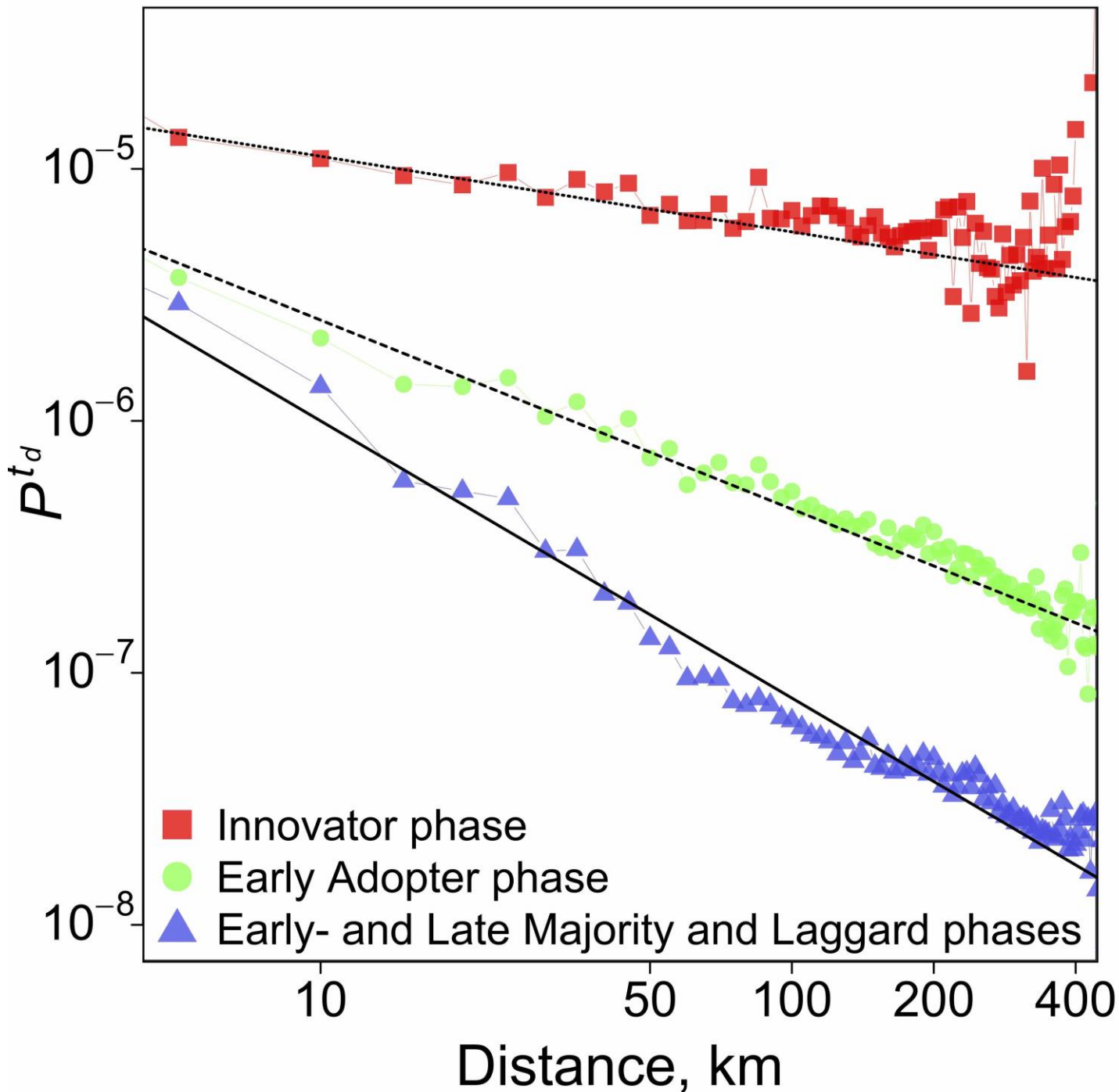
MAJORITY & LAGGARDS

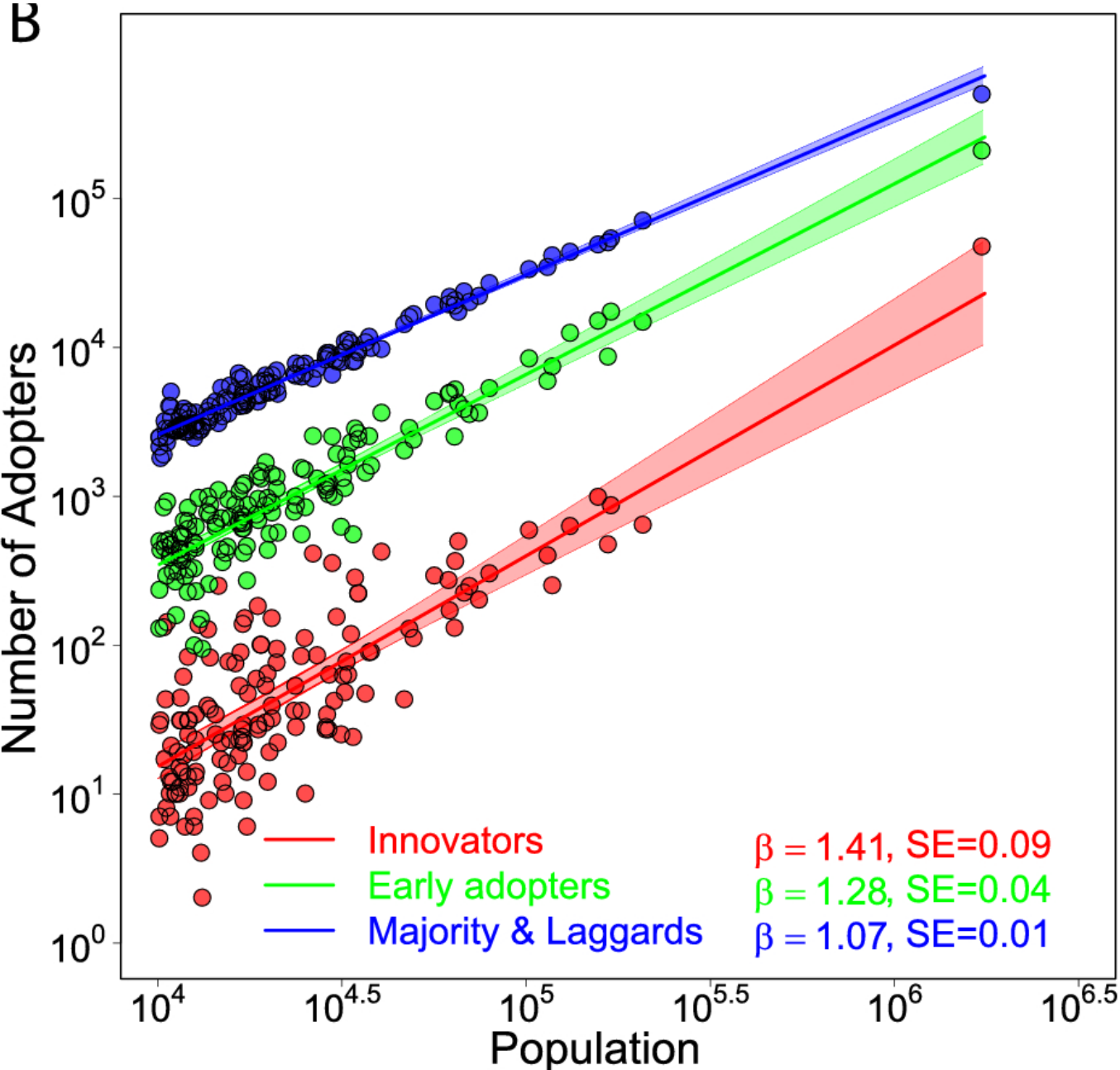


The role of distance

Distance only slightly decreases the probability of invitation in early phases

Diffusion becomes local in later phases.

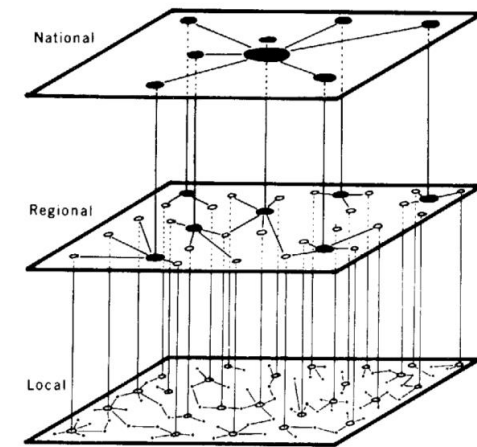




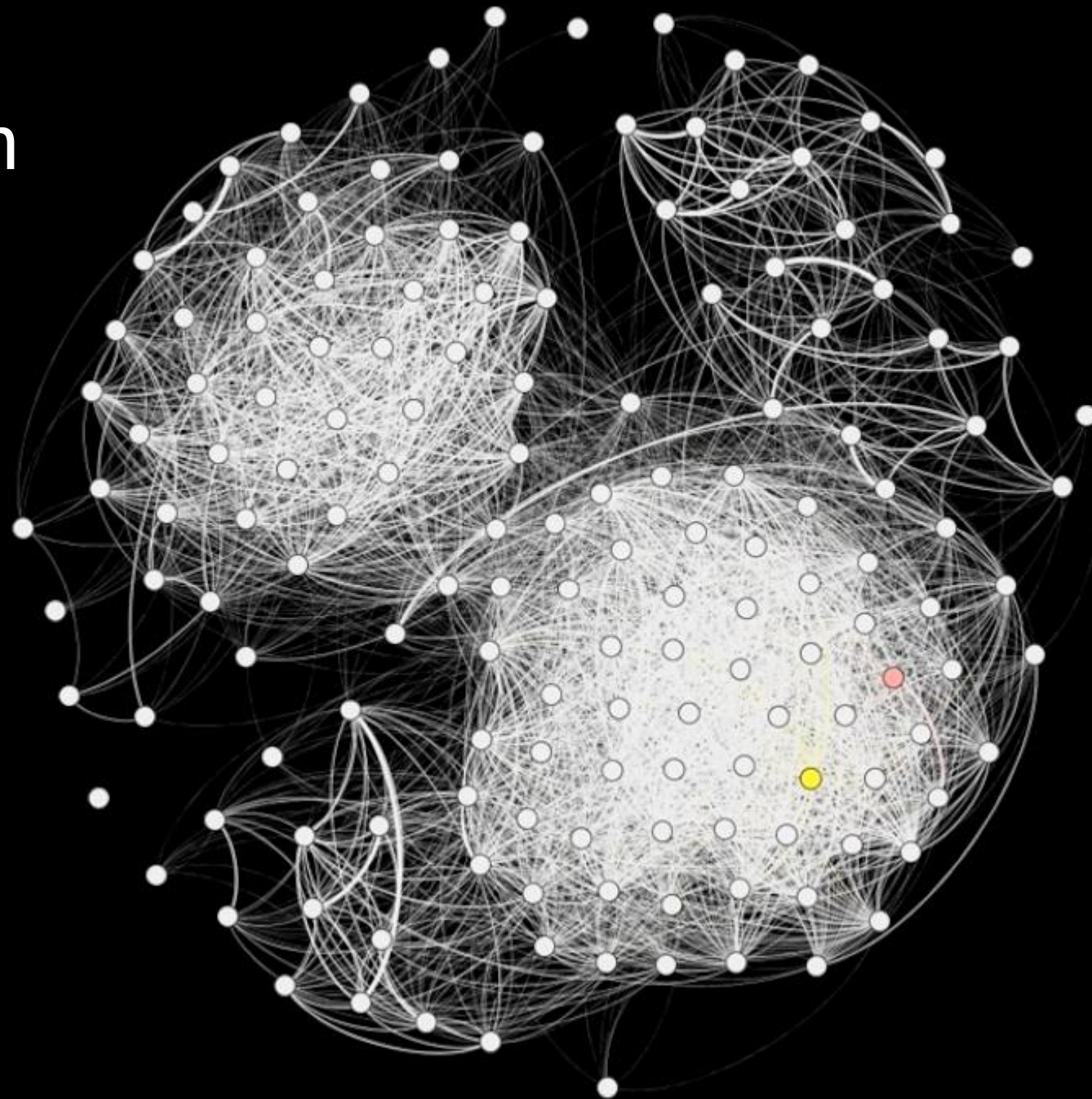
The role of size

Innovators and Early Adopters concentrate in cities.

Majority and Laggards are proportional to town size.



Diffusion in networks



Spatial wave of
adoption and
churn



The spatial life cycle hypotheses

H1: „*The churn from an innovation is early in places that have adopted it early.*”

- Declining relative advantage and social value after saturation
 - Adoption in periphery happens when centrum saturates
 - Spatial differences in perceived novelty
 - Value fades away
-
- Assumption: like adoption, churn also happens in identical spatial networks
 - Mixture of diffusion of obsolescence, new waves, local network economies

The spatial life cycle hypotheses

H2: *„Not only adoption but also churn starts in large cities.”*

- Early adoption and churn are both urban phenomena.
- Cities are more innovative and more susceptible for innovation.
- Saturation is early – obsolescence is also early.
- New products can also arrive early.

The spatial life cycle hypotheses

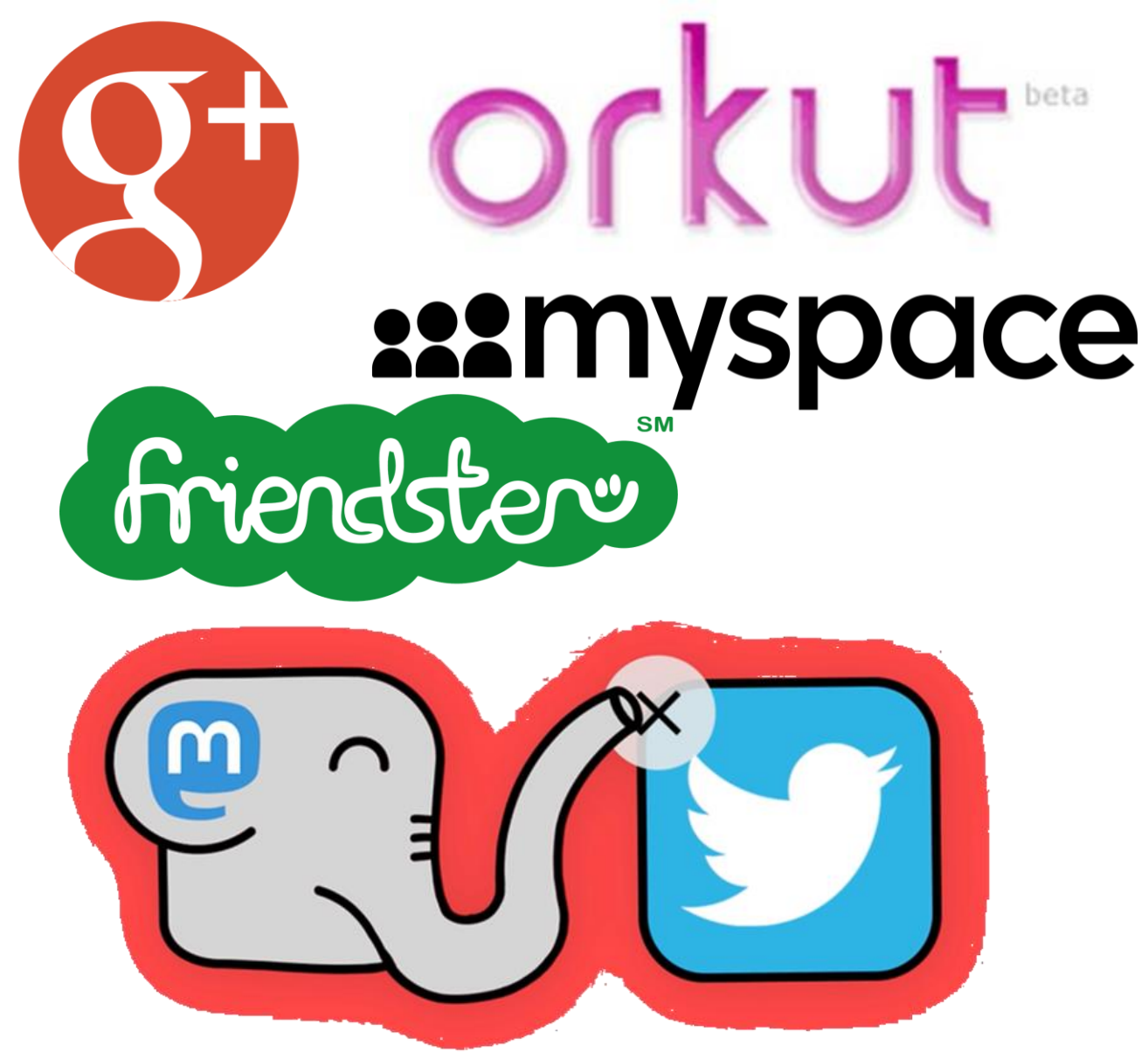
H3: *„Churn is earlier in those locations, where the adoption of the replacement innovation cycle is early.”*

- New technologies are adopted on the basis of the old one.
- Spatial diffusion is similar across waves.
- The new replaces the old.

The spatial life cycle hypotheses

H4: *„The innovation’s embeddedness in local socio-technological systems prolong its life cycle in places.”*

- Local network economies slow down churn.
- Benefits of the old technology prevail if it is embedded in local systems.
- Diffusion and churn happens in dynamic networks: local interdependencies emerge
 - Early adopters have more time to develop interdependencies
 - The networks of late adopters function more as channels of obsolescence diffusion



Not just social media:

(H2) Innovation is concentrated in cities
(Bettencourt et al. 2007)

(H3) All IT waves come from the Bay area.

(H4) Tightly connected innovation systems tend to stick to old technologies (Grabher, 1993).

This path dependency is strong in networked products.

Jane Doe 123 Main St Anywhere US 10111		Date <u>07/01/2018</u> 790
PAY TO THE ORDER OF <u>ACME Grocery Shop</u>		\$ <u>8.15</u>
<u>EIGHT AND 15/100</u> DOLLARS		
Your Bank 456 Main St Anywhere US 10111		
MEMO <u>Lunch w/Friends</u>	<u>Jane Doe</u>	
① ② ③ ④ ⑤ ⑥		⑦ ⑧ ⑨

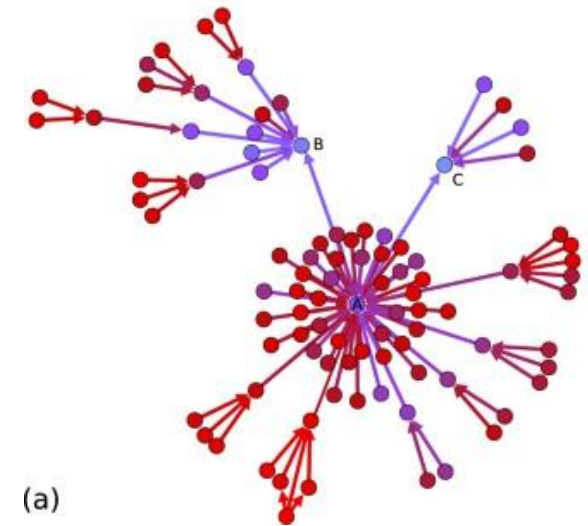
Previous research on social influence in churn

Lőrincz et al. 2019 Social Networks:

- Central individuals churn later – the network is of greater value for them.

Török and Kertész 2017, SRep

- The cascade of churn starts from the peripheral actors and their churn triggers central actors.

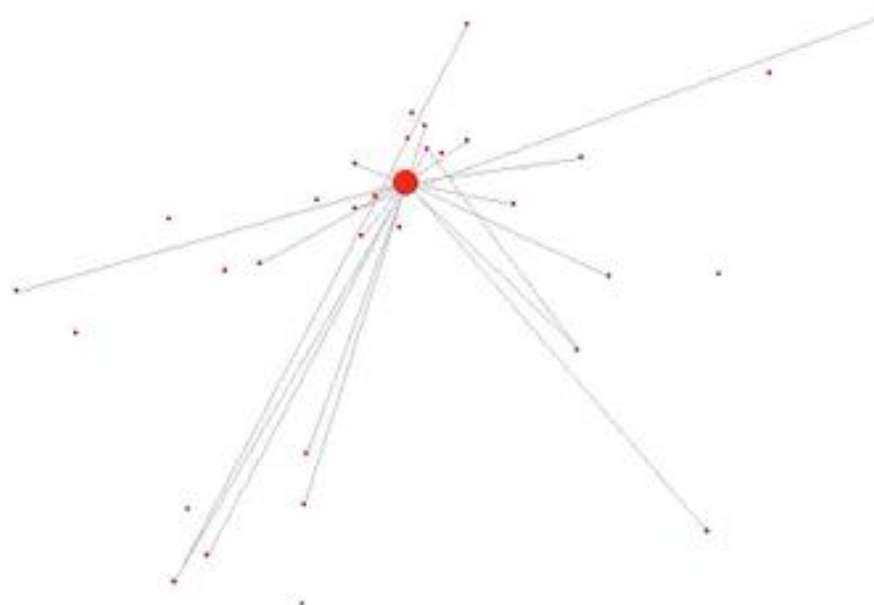
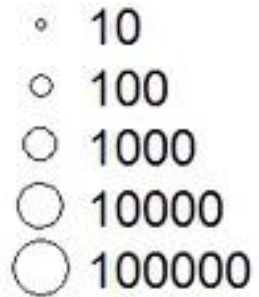


(a)

Török and Kertész (2017, Scientific Reports)

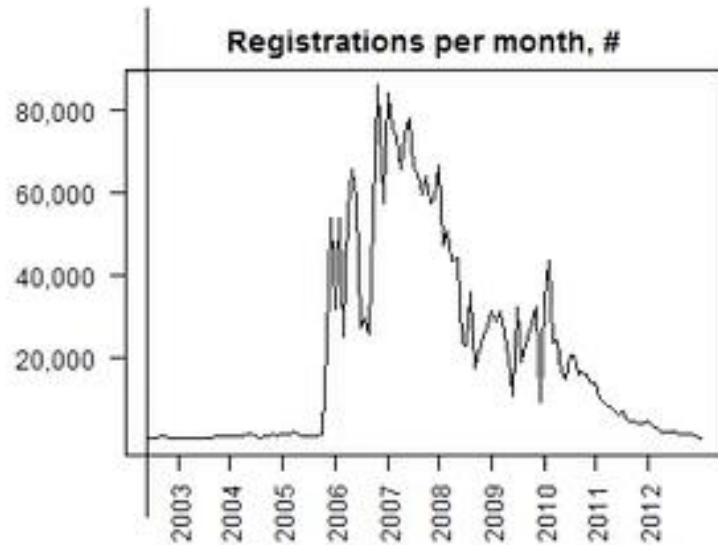
Full life-cycle of a social media platform

Registered users, #



2.8 M Users

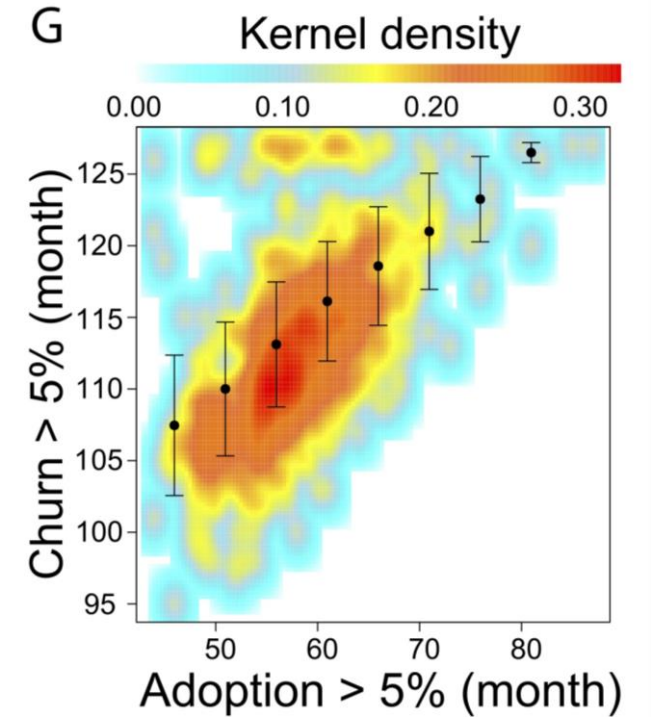
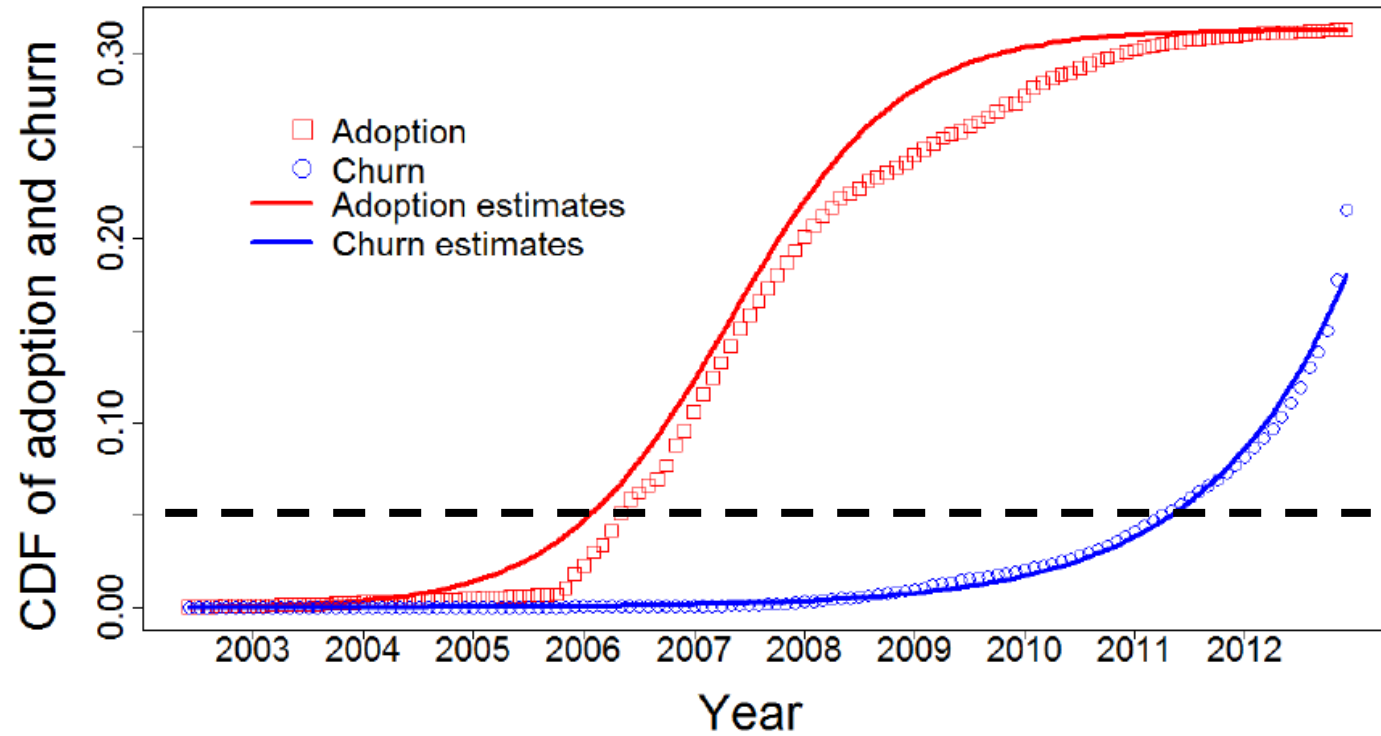
- **Location** (2557 towns),
- **Date of registration,**
- **Date of last login,**
- **Connections** (time-stamp),
- **ID of inviters**



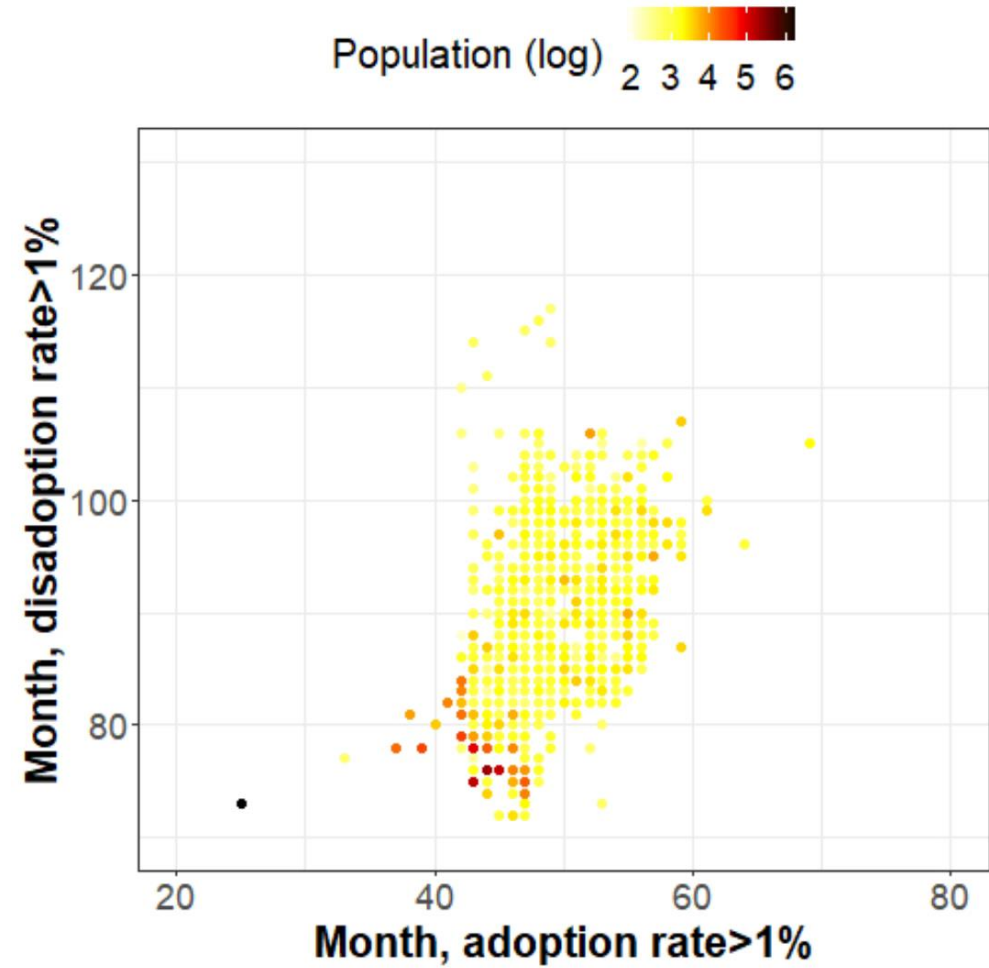
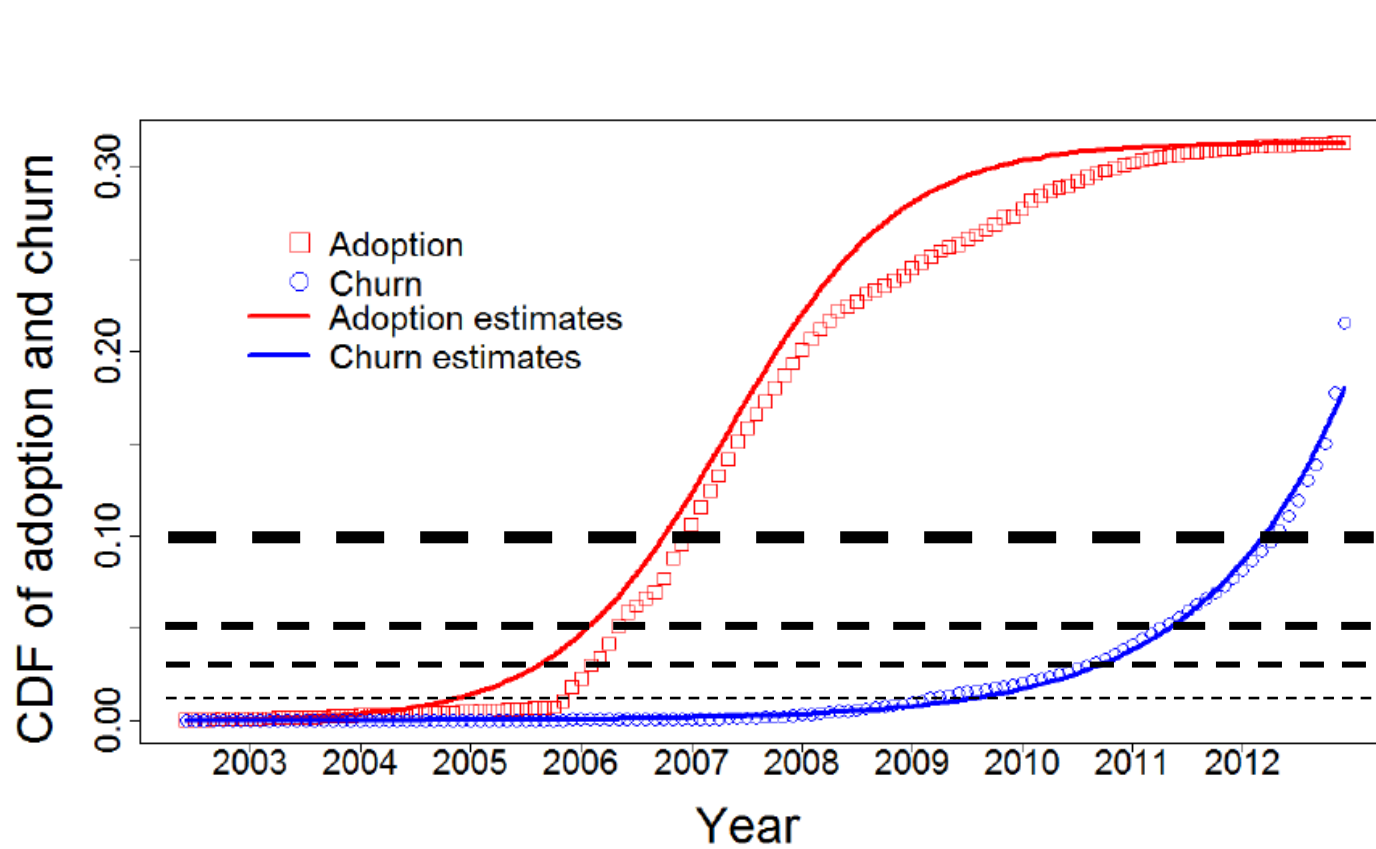
Active users, %



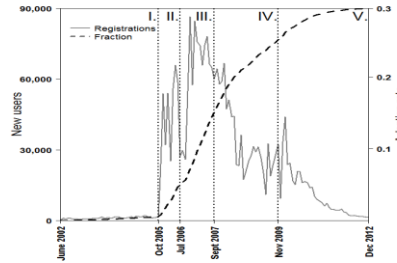
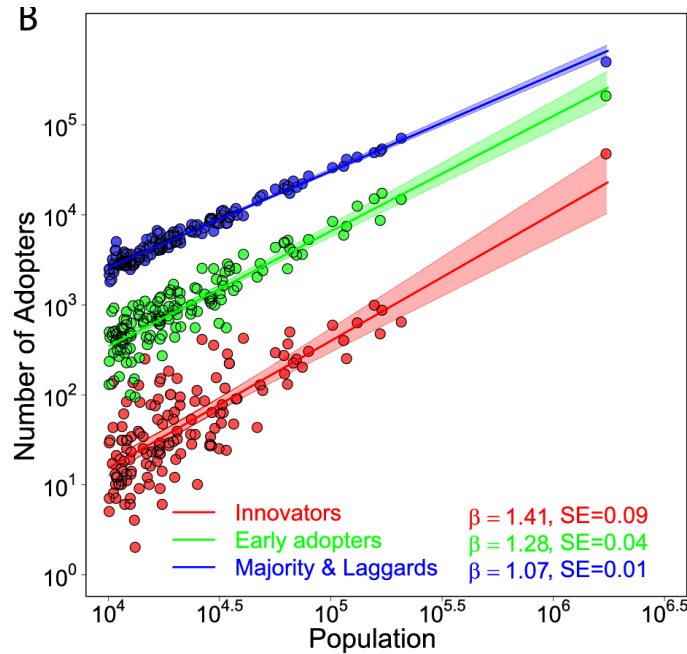
Those places churn early that adopt early



Those places churn early that adopt early

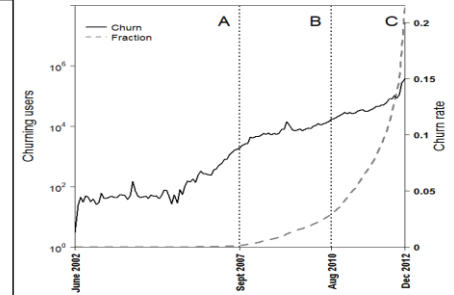
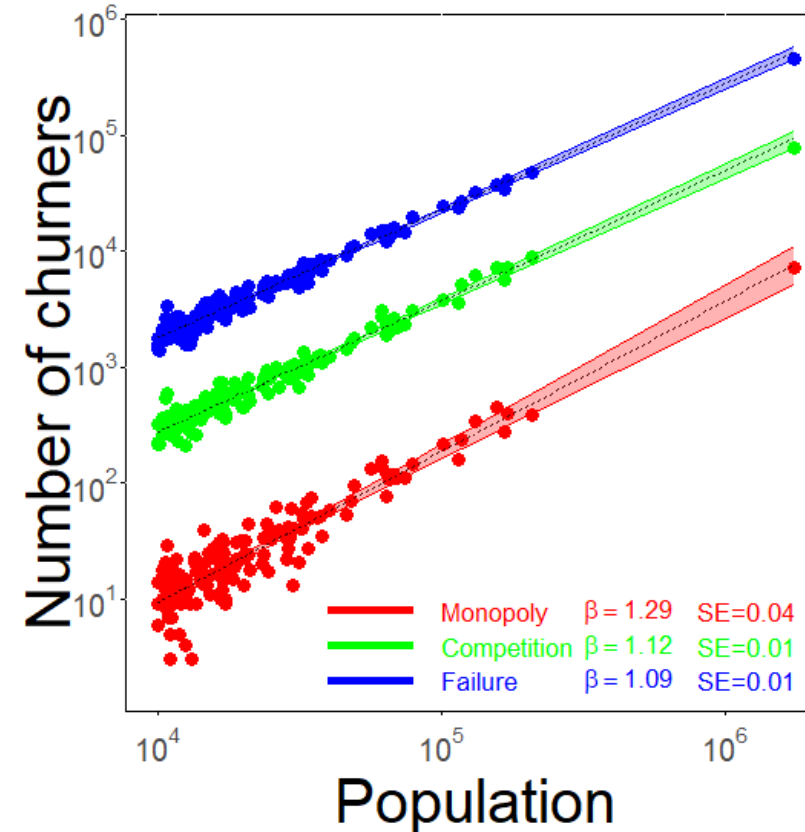


The role of city size in adoption and churn

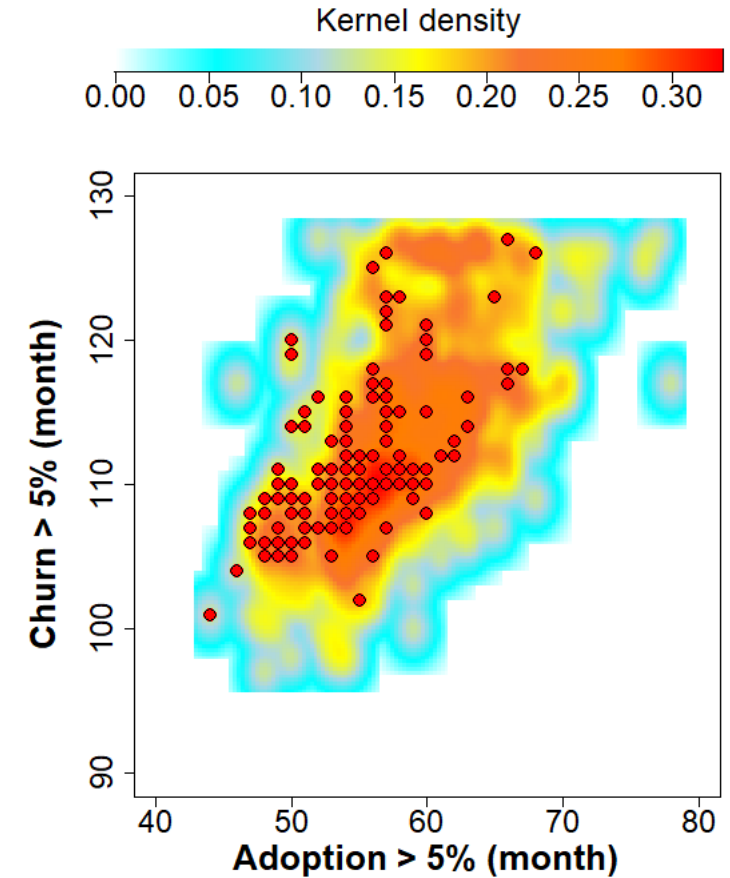
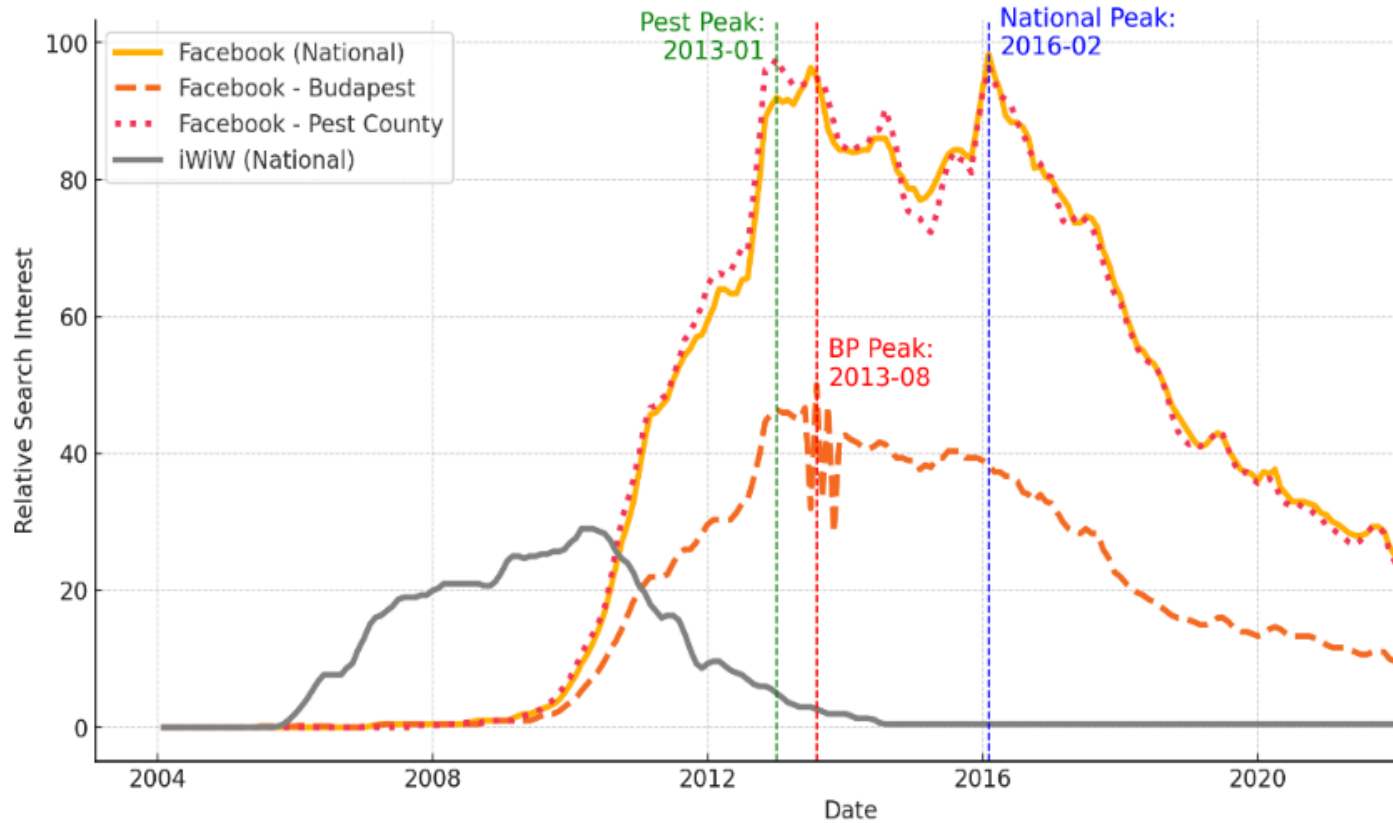


Innovators and Early Adopters
concentrate in cities.

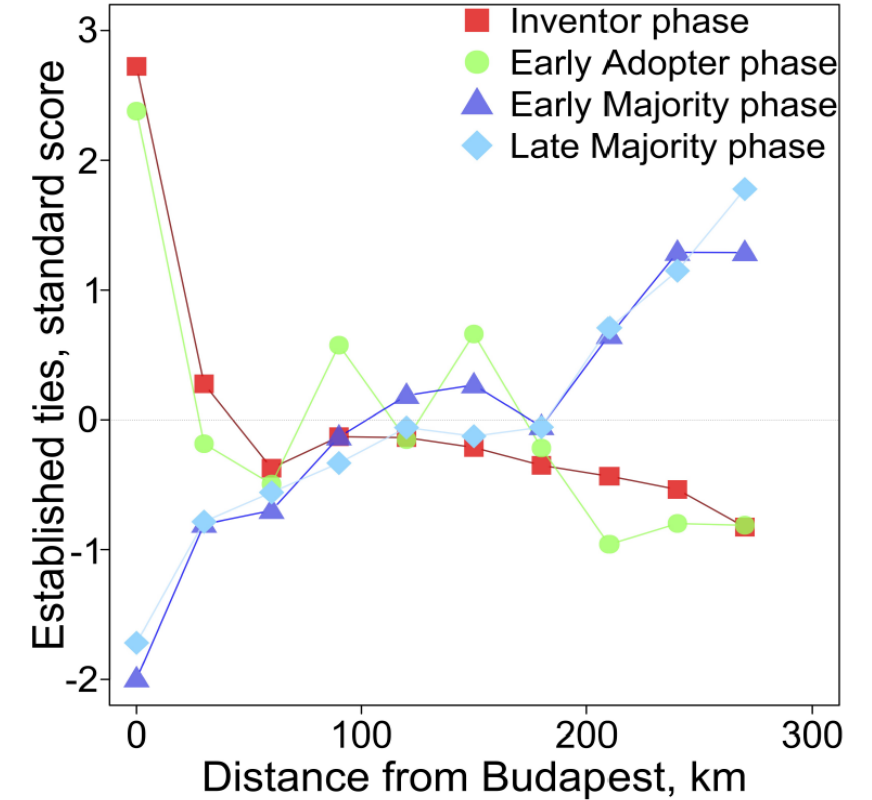
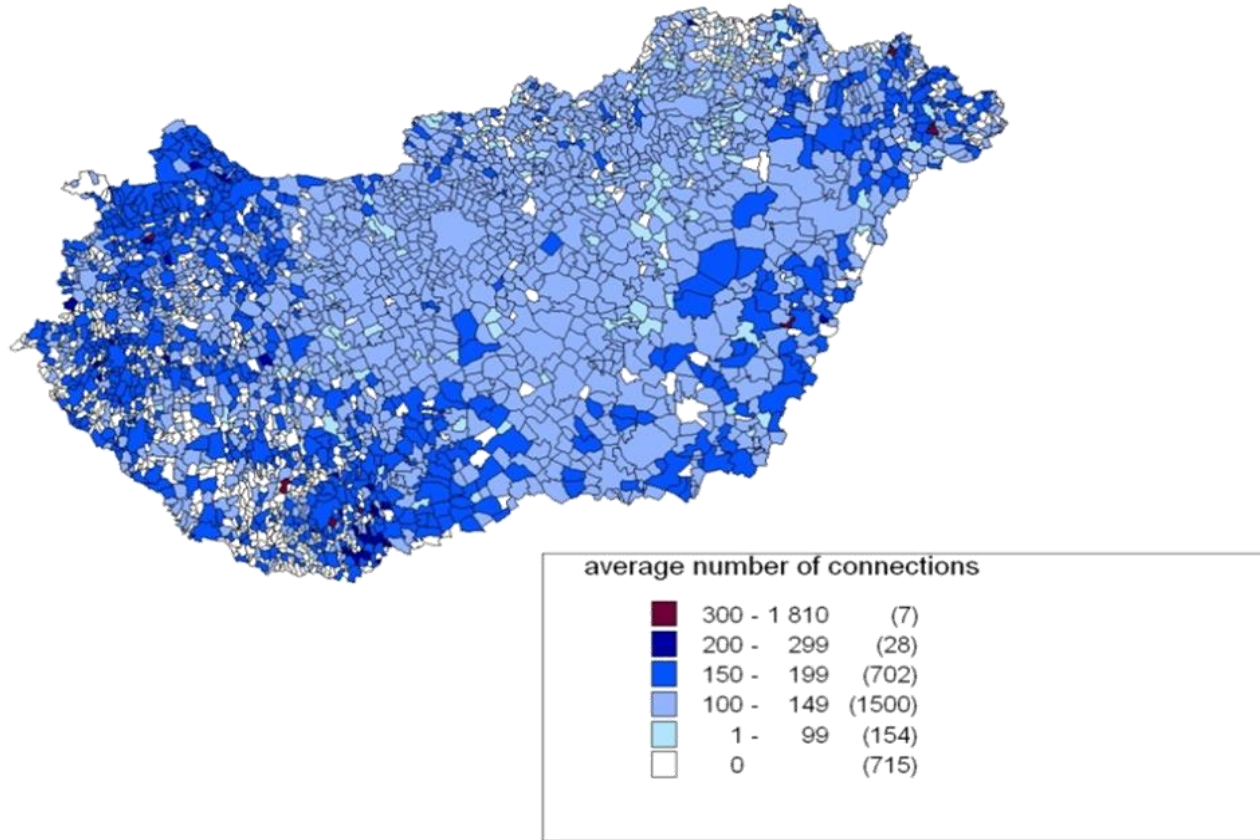
Lengyel et al. (2020, Scientific Reports)



The arrival of the new technology



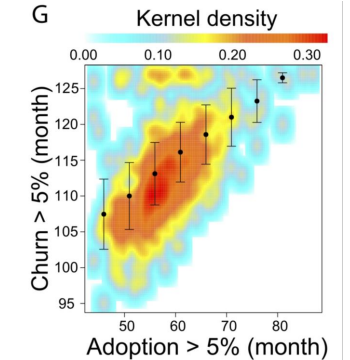
Big waves of technology usage: reporting friends



Determinants of adoption and churn dynamics, town-level OLS regression. Dependent variable: Month when town-level adoption and churn rates reach 5%. Control variables not reported: Average wage (numeric) , Ratio of taxpayers (numeric), Composite Telecommunication Index (numeric) developed in Lengyel and Jakobi (2016), University (equals 1 if the town hosts a university, else is 0), all measured in year 2008. Standard error in parentheses. Notations ***, *, * stand for statistical significance is reported at the 1%, 5% and 10% levels.

	Adoption, 5% (month)				
	(1)	(2)	(3)	(4)	(5)
Population, log	-3.994*** (0.303)			-2.479*** (0.256)	-2.260*** (0.2543)
Distance from Center, log		7.791*** (0.484)		2.749*** (0.411)	
Central Region (dummy)			-5.138*** (0.433)		-3.160*** (0.332)
Constant	72.580*** (1.035)	43.389*** (0.982)	59.732*** (0.157)	75.049*** (1.593)	81.781*** (1.265)
Town Controls	NO	NO	NO	YES	YES
N	1,324	1,322	1,324	1,322	1,324
R2	0.116	0.164	0.096	0.549	0.564
F	174.276***	259.242***	140.854***	266.774***	284.192***

	Churn, 5% (month)						
	(0)	(1)	(2)	(3)	(4)	(5)	(6)
Month of adoption 5%	0.865*** (0.035)						0.984* (0.050)
Population, log		-7.601*** (0.424)			-7.669*** (0.521)	-7.591*** (0.525)	-5.342* (0.458)
Distance from Center, log			2.945*** (0.717)		-2.158*** (0.792)		-3.461* (0.676)
Central Region (dummy)				-2.793*** (0.612)		-0.391 (0.613)	
Constant	62.961*** (2.033)	139.956*** (1.499)	107.409*** (1.446)	113.702*** (0.238)	154.070*** (3.579)	147.757*** (2.942)	76.562* (7.056)
Town Controls	NO	NO	NO	NO	YES	YES	YES
N	1,021	1,021	1,021	1,021	1,021	1,021	1,021
R2	0.377	0.240	0.016	0.020	0.302	0.297	0.496
F	617.344***	321.966***	16.873***	20.814***	73.155***	71.490***	142.385



Social connections and churn dynamics, town-level OLS regression. Dependent variable: Month when town-level churn rates reach 5%. Control variables not reported: Average wage (numeric) , Ratio of taxpayers (numeric), Composite Telecommunication Index (numeric) developed in Lengyel and Jakobi (2016), University (equals 1 if the town hosts a university, else is 0), all measured in year 2008. Standard error in parentheses. Notations ***, *, * stand for statistical significance is reported at the 1%, 5% and 10% levels.

	Churn, 5% (month)		
	(1)	(2)	(3)
Month of adoption 5%	0.838*** (0.037)	0.933*** (0.041)	1.169*** (0.054)
Population, log	-5.102*** (0.384)	-5.306*** (0.412)	-5.630*** (0.461)
Avg. degree	-0.045*** (0.005)		
Avg. degree, Early Majority		0.059*** (0.009)	0.069*** (0.009)
Avg. degree, Late Majority		-0.085*** (0.008)	-0.081*** (0.008)
Avg. Degree, Laggards		0.000*** (0.000)	0.000*** (0.000)
Constant	91.058*** (2.831)	81.273*** (3.384)	58.084*** (5.401)
Town Controls	NO	NO	YES
N	1,021	1,021	1,021
R2	0.492	0.512	0.535
F	328.188***	213.014***	129.173***

Further steps: generalization and synergies

Different contexts

- Web technologies
- Technologies from patent (and trademark) data
- Scientific terms, concepts, keywords in publications
- Topics and expressions in news

Mechanisms

- Better identification of substituting technologies

Theory and models

- Recovery like in the network epidemiology models
- Regional economics and economic development